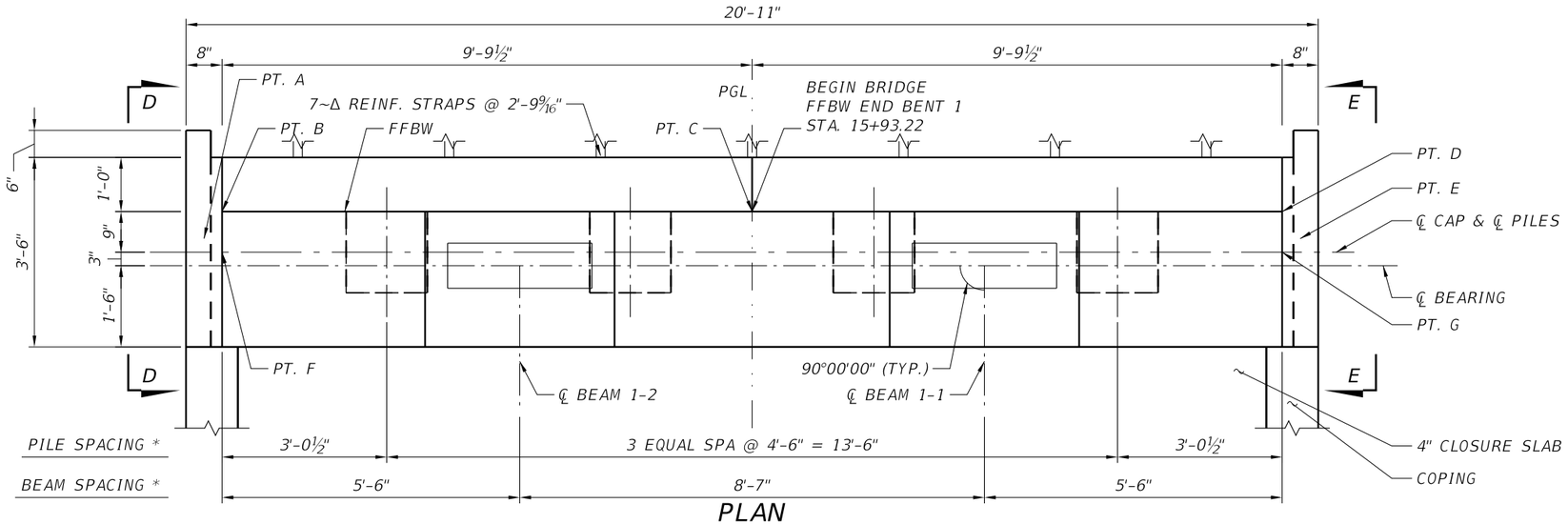
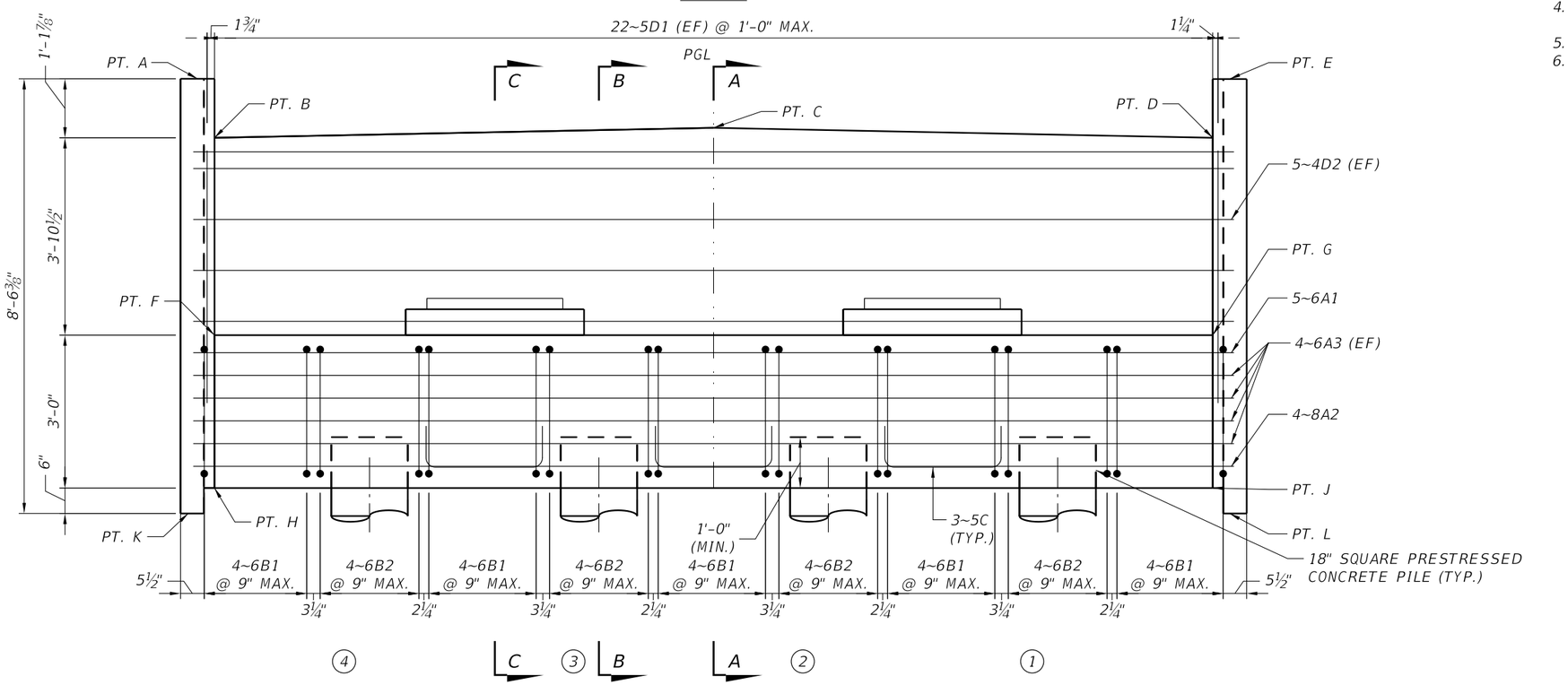


PLAN



* DIMENSIONS MEASURED ALONG FFBW

ELEVATION

(LOOKING BACK STATION)

3/8" = 1'-0"

TABLE OF ELEVATIONS	
POINT	ELEVATION
A	48.269
B	47.111
C	47.307
D	47.111
E	48.265
F	43.238
G	43.238
H	40.238
J	40.238
K	39.738
L	39.738

PILE CUTOFF ELEVATIONS		
PILE NO.	CUTOFF EL.	PGL OFFSET
1	41.2	-6'-9"
2	41.2	-2'-3"
3	41.2	2'-3"
4	41.2	6'-9"

DIRECTION OF STATIONING

NOTES:

1. STIRRUP SPACING IS DIMENSIONED TO BAR CENTERLINES.
2. REINFORCEMENT IS SHOWN REPRESENTATIVELY: THE FIRST AND LAST BAR OF EACH RUN ARE DRAWN.
3. FOR SECTIONS A-A AND B-B AND DETAILS "A" AND "B", SEE SHEET B4-2. FOR SECTION C-C AND VIEWS D-D AND E-E, SEE SHEET B4-3. FOR SECTIONS F-F AND G-G AND PEDESTAL DETAILS, SEE SHEET B4-4.
4. FOR THE SOIL REINFORCEMENT (Δ) LEGEND, SEE SHEET B4-2.
5. FOR REINFORCING BAR LIST, SEE SHEET B4-16.
6. PILE OFFSETS ARE FROM THE PGL, + TO THE RIGHT LOOKING UP-STATION. CUTOFF ELEVATIONS ARE TO THE NEAREST 0.1 FT.

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS					
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DRAWN BY:	
CHECKED BY:	
DESIGNED BY:	
CHECKED BY:	

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

ROAD NO.	COUNTY	FINANCIAL PROJECT ID

SHEET TITLE: END BENT 1

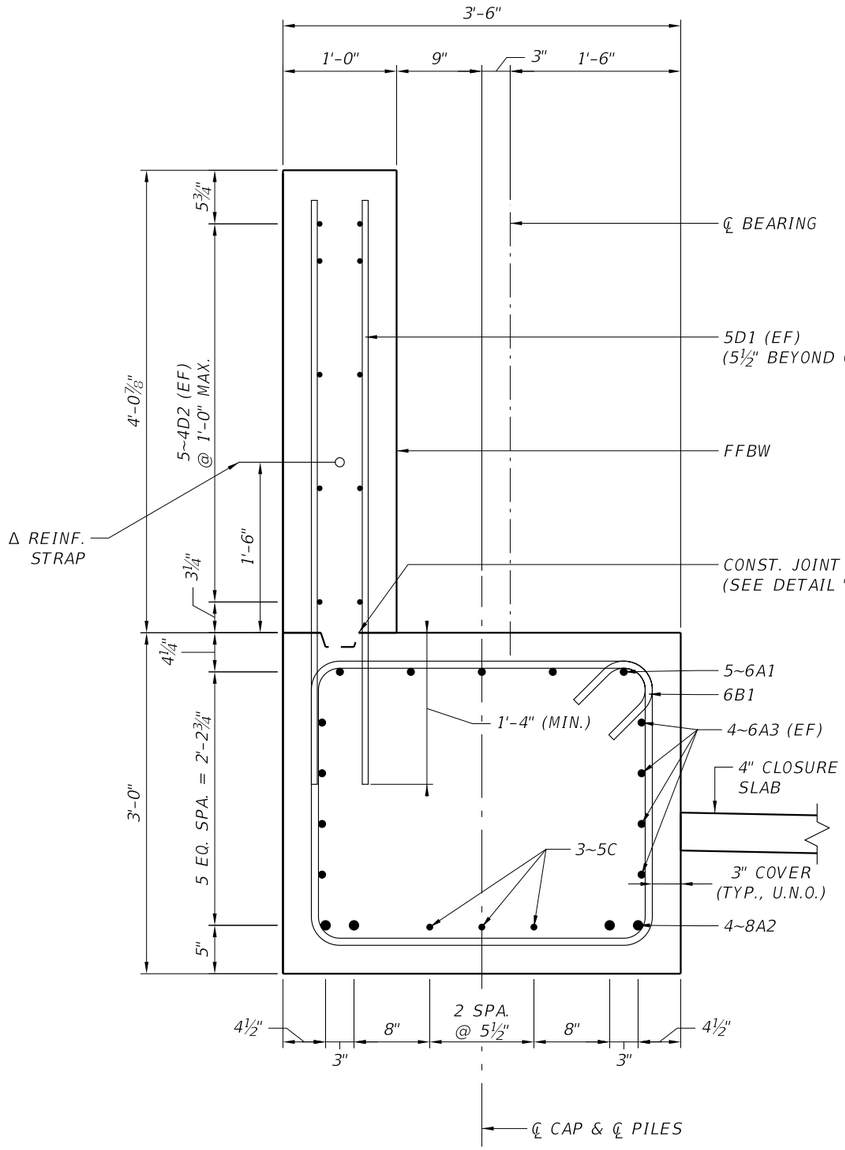
PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE - B4

REF. DWG. NO.

SHEET NO. B4-1

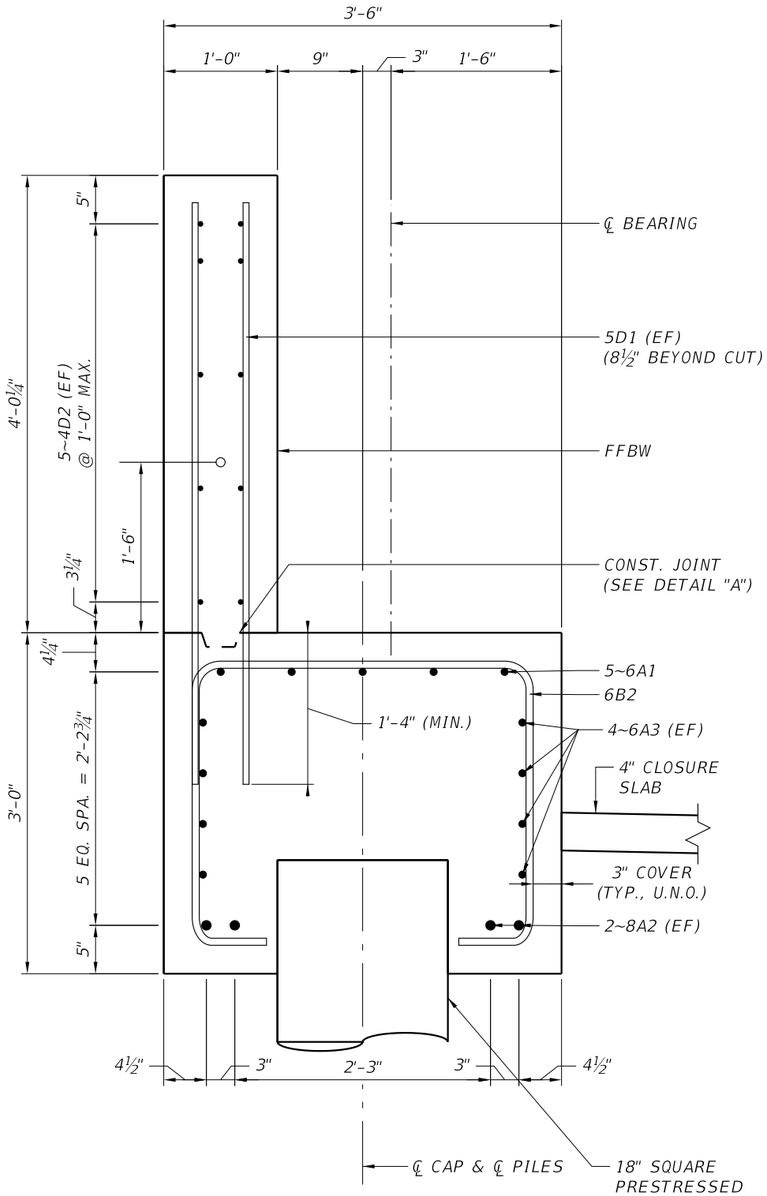
THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)



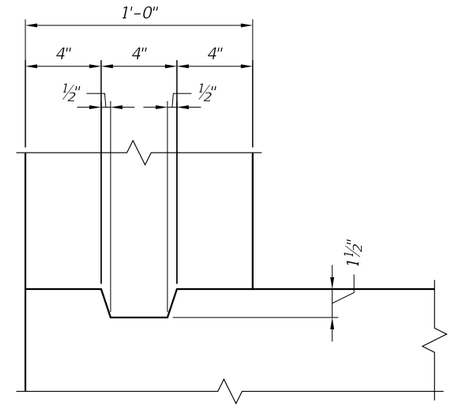
SECTION A-A

(SECTION BETWEEN PILES)



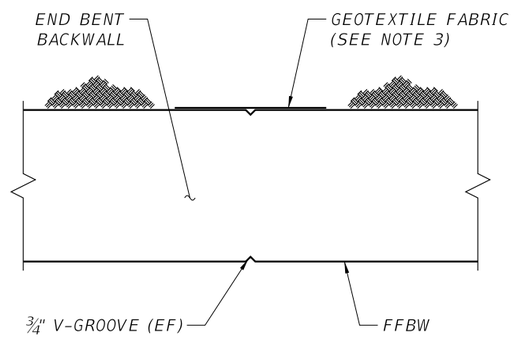
SECTION B-B

(SECTION AT PILES)



DETAIL "A"

(CONSTRUCTION JOINT)



DETAIL "B"

(END BENT BACKWALL V-GROOVE)

LEGEND:

Δ REINFORCING STRAPS AND ATTACHMENTS FOR THE END BENT SHALL BE DESIGNED AND FURNISHED BY THE PROPRIETARY WALL PROVIDER. THE STRAPS SHALL BE DESIGNED TO RESIST A SERVICE LOAD OF 3.09 K/FT AND A FACTORED LOAD OF 3.74 K/FT (STRENGTH I) PERPENDICULAR TO THE END BENT BACKWALL.

NOTES:

1. FOR LOCATION OF SECTIONS, SEE SHEET B4-1.
2. ALL BARS ARE SHOWN IN SECTIONS; BARS IN THE SECTION PLANE ARE THE NEAREST BAR OF THEIR SERIES WITHIN 1'-3" BEYOND THE CUT.
3. ATTACH TYPE D-5 (SPECIFICATIONS SECTION 985) GEOTEXTILE FABRIC, 1'-0" WIDE AND FULL HEIGHT OF FILL, TO THE BACK OF BACKWALL AT V-GROOVE LOCATIONS WITH AN ADHESIVE APPROVED BY THE ENGINEER.
4. MODEL CHECK (REMOVE BEFORE ISSUE): DETAILS "A" AND "B" ARE SCHEMATIC; THE CONSTRUCTION JOINT KEY AND THE V-GROOVE LOCATIONS ARE NOT IN THE CAD MODEL.
5. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

¾" = 1'-0"

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS					
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DRAWN BY:
CHECKED BY:
DESIGNED BY:
CHECKED BY:

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

ROAD NO.	COUNTY	FINANCIAL PROJECT ID

SHEET TITLE: END BENT 1 DETAILS (1 OF 3)

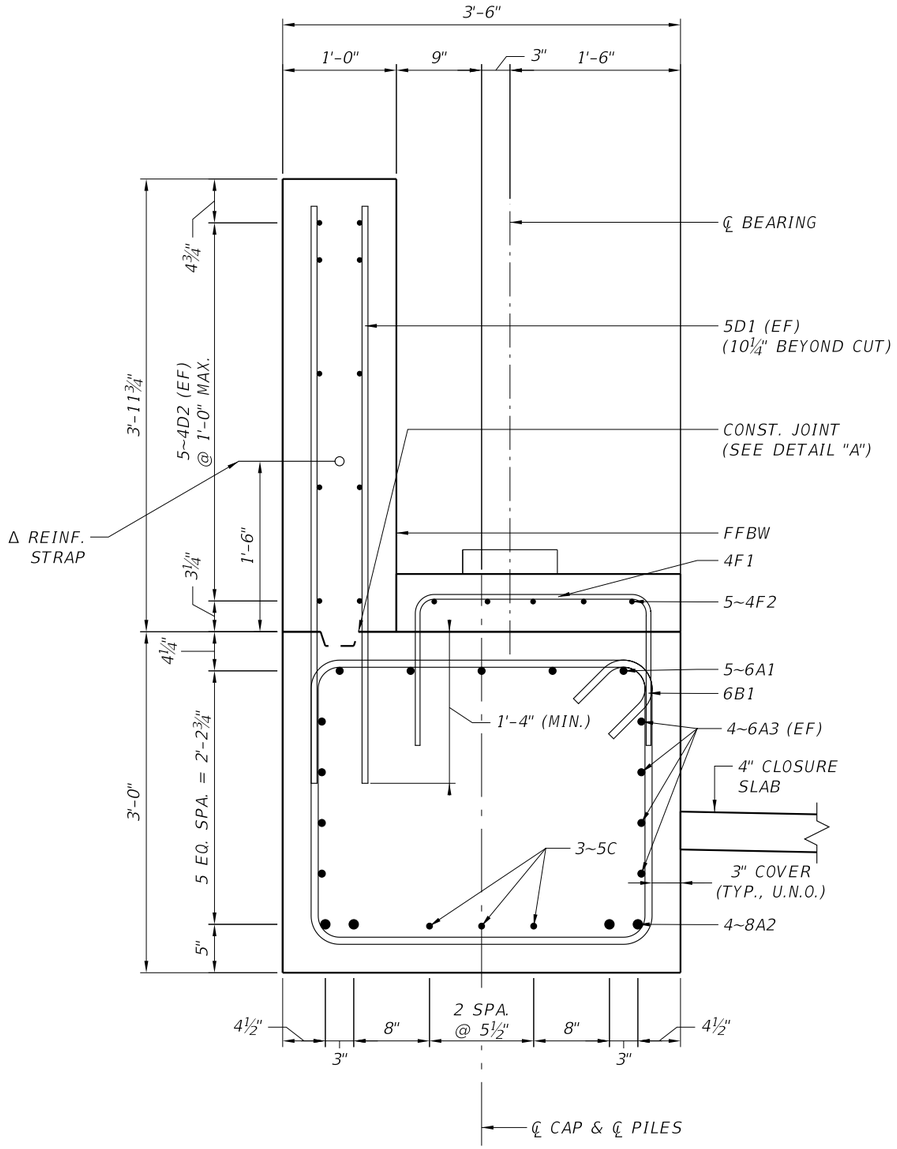
PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE – B4

REF. DWG. NO.

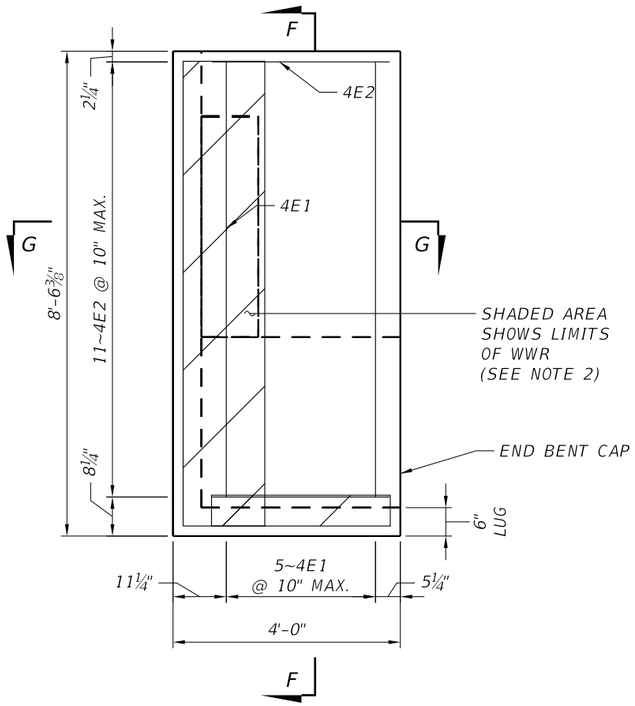
SHEET NO. B4-2

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

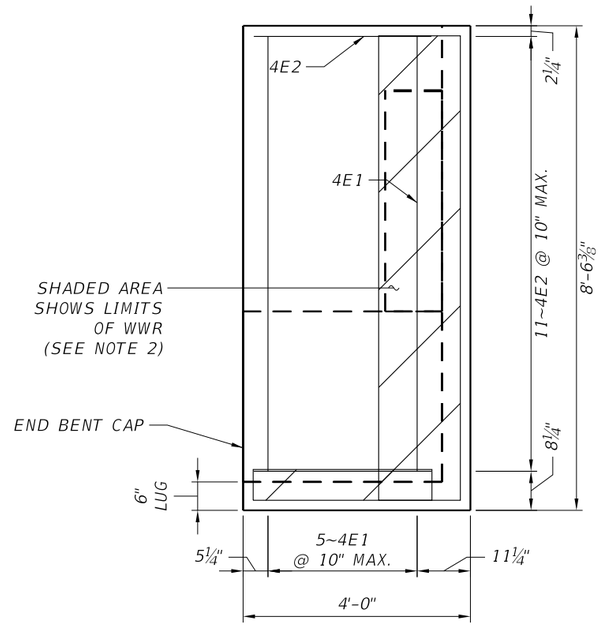
FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)



SECTION C-C
(SECTION AT CENTERLINE OF BEAM)



VIEW D-D
(LEFT CHEEK WALL)



VIEW E-E
(RIGHT CHEEK WALL)

NOTES:

1. FOR LOCATION OF SECTION C-C AND VIEWS D-D AND E-E, SEE SHEET B4-1.
2. WELDED WIRE REINFORCEMENT (WWR) 4x4-W1.4xW1.4 IN THE CHEEK LUGS SHALL BE TIED TO THE END BENT REINFORCING STEEL.
3. FOR CHEEK TOP ELEVATIONS, SEE THE TABLE OF ELEVATIONS, SHEET B4-1.
4. SECTIONS F-F AND G-G ARE LOCATED IN VIEW D-D; FOR SECTIONS F-F AND G-G, SEE SHEET B4-4.
5. FOR THE Δ REINFORCING STRAP LEGEND, SEE SHEET B4-2.
6. MODEL CHECK (REMOVE BEFORE ISSUE): THE BACKWALL TOP IS LEVEL ACROSS THE WALL THICKNESS IN THE CAD MODEL (ITS SLOPE RUNS ALONG THE BENT TO THE CROWN ONLY); THE SECTIONS SHOW THE MODEL. ADD THE CROSS SLOPE TO THE MODEL.
7. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

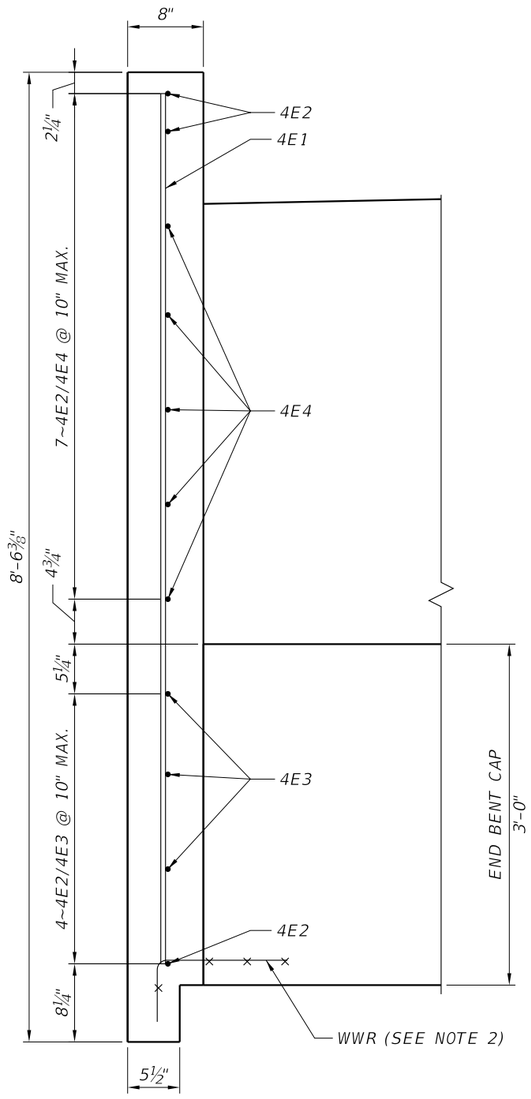
3/4" – 1'-0" 3/8" – 1'-0"

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS						ENGINEER OF RECORD:	DRAWN BY:	STATE OF FLORIDA DEPARTMENT OF TRANSPORTATION			SHEET TITLE: END BENT 1 DETAILS (2 OF 3)	REF. DWG. NO.
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION		CHECKED BY:					
							DESIGNED BY:	ROAD NO.	COUNTY	FINANCIAL PROJECT ID	PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE – B4	SHEET NO. B4-3
							CHECKED BY:					

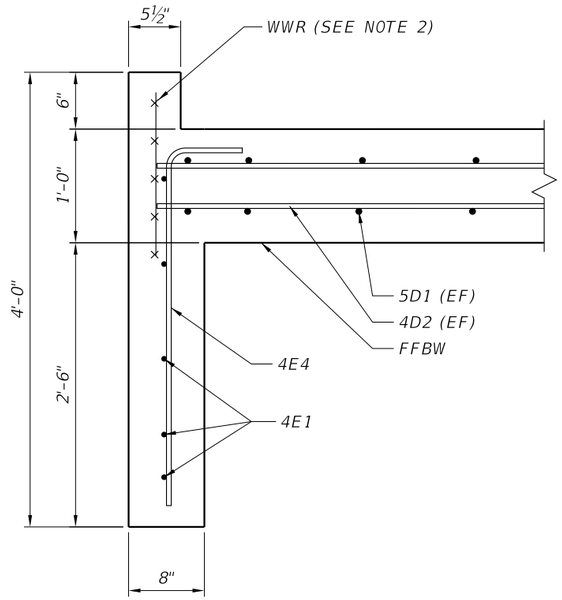
FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.



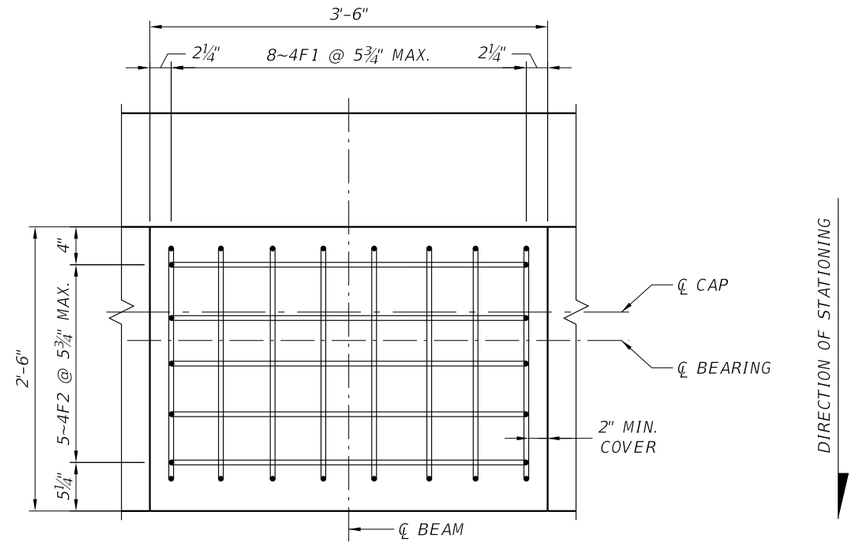
SECTION F-F

(THROUGH CHEEK WALL)



SECTION G-G

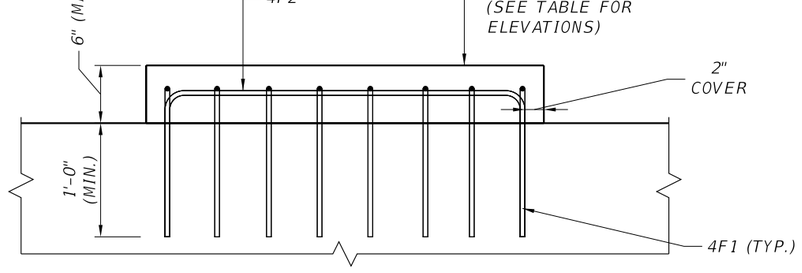
(THROUGH CHEEK WALL AND BACKWALL)



PEDESTAL PLAN

(BEAM LINE 1 SHOWN; BAR SPACING MAY BE ADJUSTED TO CLEAR THE CAP REINFORCEMENT)

TOP OF PEDESTAL ELEVATIONS	
BEAM NO.	ELEVATION (FEET)
1-1	43.746
1-2	43.746



PEDESTAL ELEVATION

NOTES:

1. FOR LOCATION OF SECTIONS F-F AND G-G, SEE VIEW D-D, SHEET B4-3; SECTIONS F-F AND G-G SHOW THE LEFT CHEEK WALL, THE RIGHT CHEEK WALL IS SIMILAR (MIRRORED).
2. WELDED WIRE REINFORCEMENT (WWR) 4x4-W1.4xW1.4 IN THE CHEEK LUGS SHALL BE TIED TO THE END BENT REINFORCING STEEL.
3. CAP AND BACKWALL REINFORCEMENT IS NOT SHOWN IN SECTION F-F FOR CLARITY.
4. PEDESTALS AT THE OTHER BEAM LINES ARE SIMILAR.
5. PEDESTALS ARE CAST ON THE CAP; THE CAP TOP IS A CONSTRUCTION JOINT.
6. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

3/4" = 1'-0"

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS					
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DRAWN BY:
CHECKED BY:
DESIGNED BY:
CHECKED BY:

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

ROAD NO.	COUNTY	FINANCIAL PROJECT ID

SHEET TITLE: END BENT 1 DETAILS (3 OF 3)

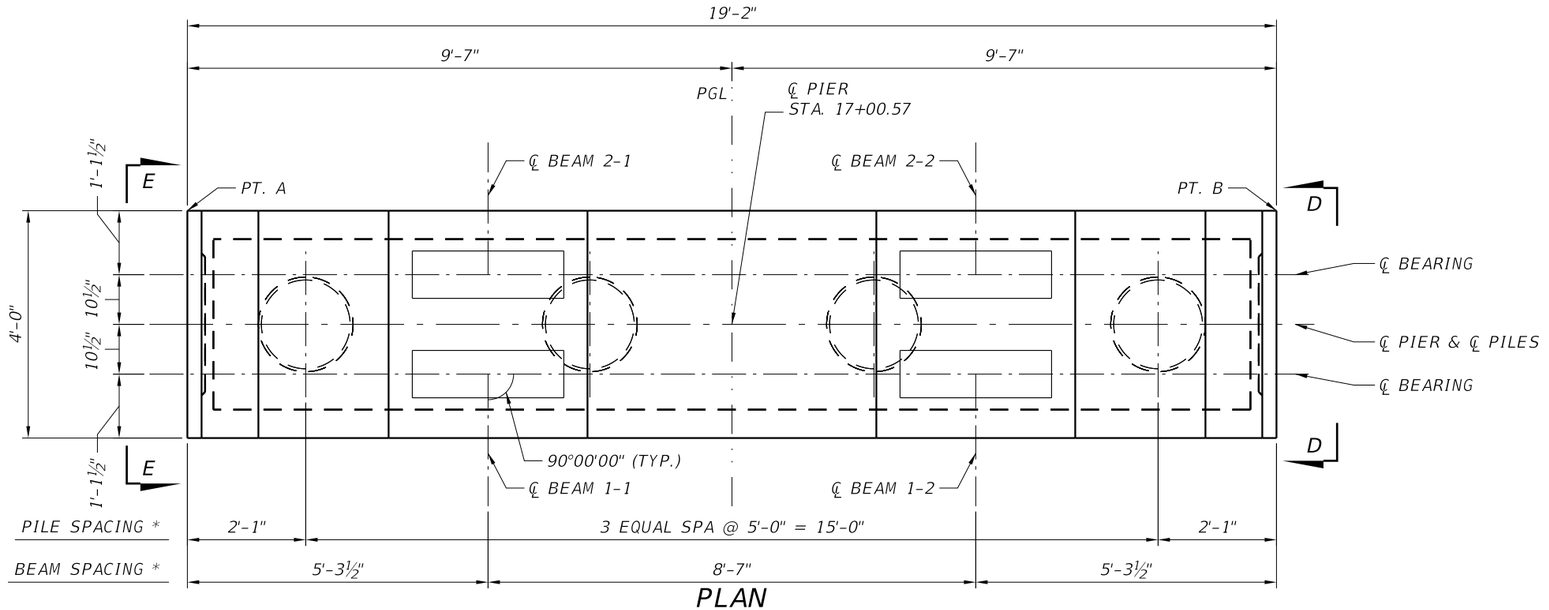
PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE - B4

REF. DWG. NO.

SHEET NO. B4-4

FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.



PLAN

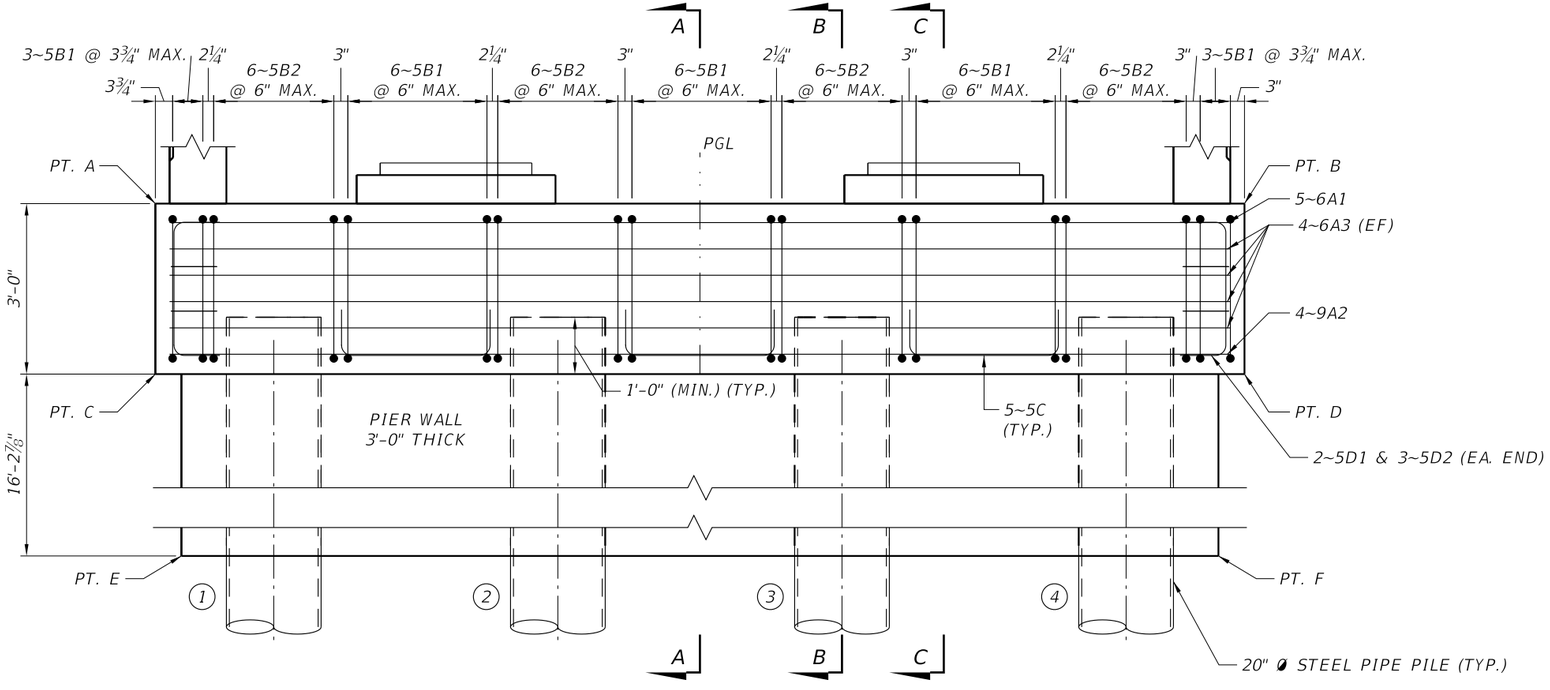
DIRECTION OF STATIONING

TABLE OF ELEVATIONS	
POINT	ELEVATION
A	43.765
B	43.765
C	40.765
D	40.765
E	24.526
F	24.526

PILE CUTOFF ELEVATIONS		
PILE NO.	CUTOFF EL.	PGL OFFSET
1	41.8	-7'-6"
2	41.8	-2'-6"
3	41.8	2'-6"
4	41.8	7'-6"

NOTES:

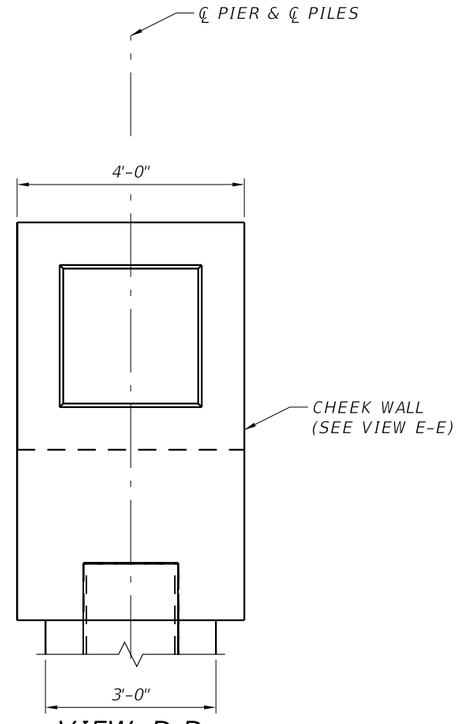
1. STIRRUP SPACING IS DIMENSIONED TO BAR CENTERLINES.
2. REINFORCEMENT IS SHOWN REPRESENTATIVELY: THE FIRST AND LAST BAR OF EACH RUN ARE DRAWN.
3. FOR SECTIONS A-A, B-B AND C-C, SEE SHEET B4-6. FOR PEDESTAL DETAILS, SEE SHEET B4-7. FOR VIEW E-E AND THE CHEEK WALL SECTIONS F-F AND G-G, SEE SHEET B4-8. FOR LOCATION OF VIEWS D-D AND E-E, SEE PLAN.
4. THE CHEEK WALLS ARE SHOWN BROKEN IN THE ELEVATION AND WHOLE IN VIEW D-D AND ON SHEET B4-8.
5. FOR THE PIER WALL REINFORCEMENT AND THE ANNULAR PILE RING PLATES, SEE SHEETS B4-9 TO B4-11.
6. FOR REINFORCING BAR LIST, SEE SHEET B4-16.
7. PILE OFFSETS ARE FROM THE PGL, + TO THE RIGHT LOOKING UP-STATION. CUTOFF ELEVATIONS ARE TO THE NEAREST 0.1 FT.



* DIMENSIONS MEASURED ALONG ℄ PIER

ELEVATION

(LOOKING AHEAD STATION)



VIEW D-D

(END VIEW)

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS					
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DRAWN BY:
CHECKED BY:
DESIGNED BY:
CHECKED BY:

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

ROAD NO.	COUNTY	FINANCIAL PROJECT ID

SHEET TITLE: PIER

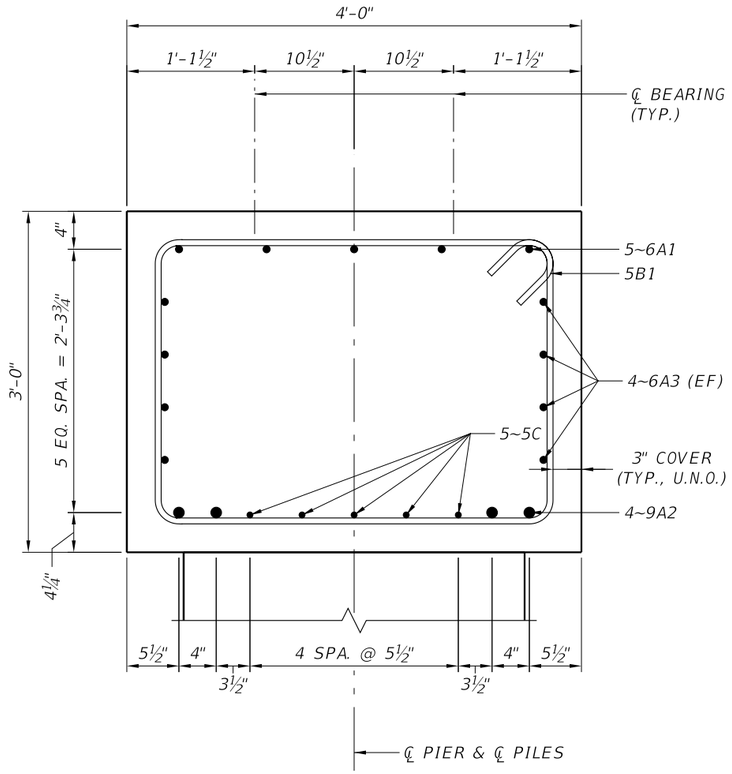
PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE - B4

REF. DWG. NO.

SHEET NO. B4-5

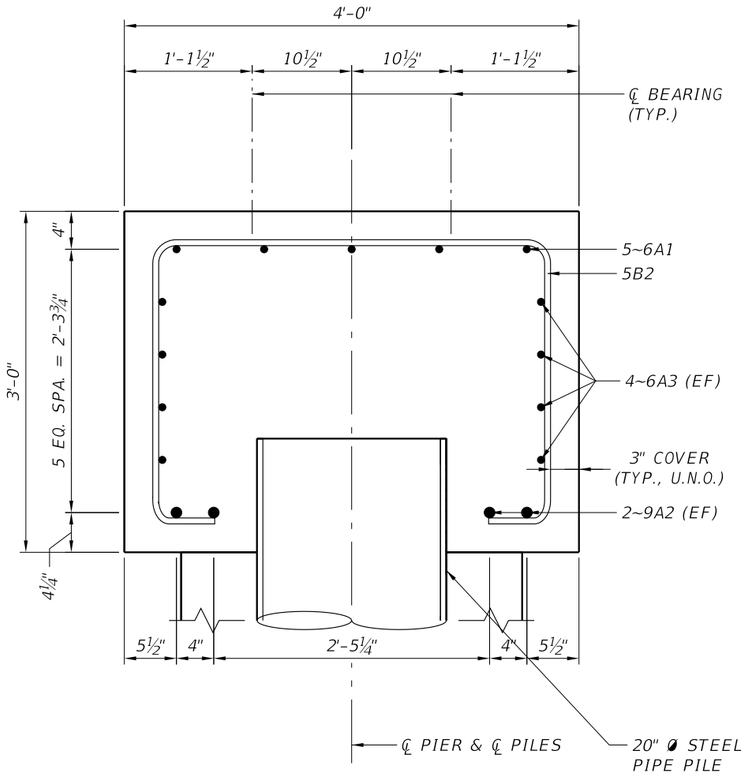
THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)



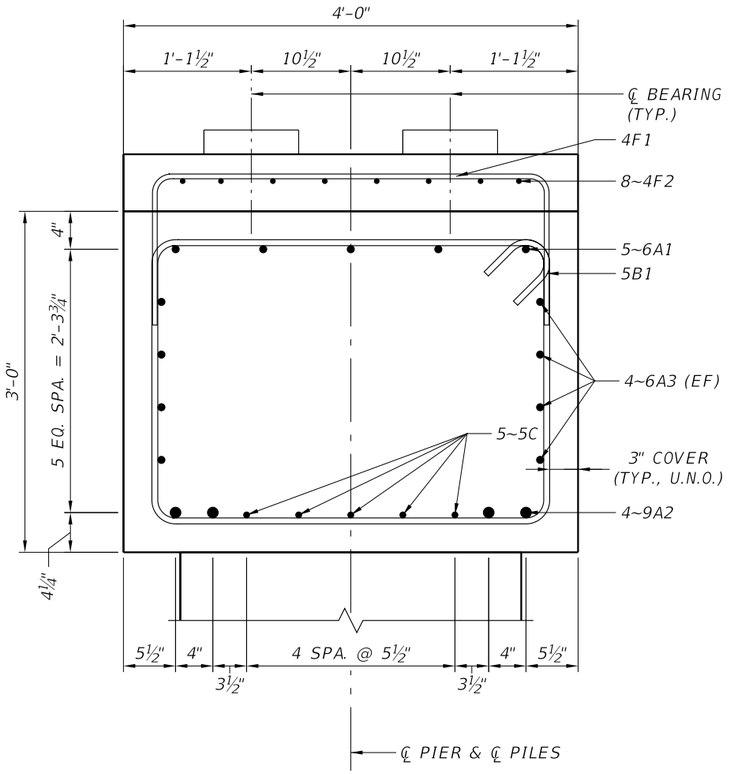
SECTION A-A

(SECTION BETWEEN PILES)



SECTION B-B

(SECTION AT PILES)



SECTION C-C

(SECTION AT CENTERLINE OF BEAMS)

NOTES:

1. FOR LOCATION OF SECTIONS, SEE SHEET B4-5.
2. ALL BARS ARE SHOWN IN SECTIONS; BARS IN THE SECTION PLANE ARE THE NEAREST BAR OF THEIR SERIES WITHIN 1'-3" BEYOND THE CUT.
3. FOR PEDESTAL DETAILS, SEE SHEET B4-7.
4. THE PIER WALL REINFORCEMENT IS NOT SHOWN IN THE WALL STUBS; SEE SHEETS B4-9, B4-10 AND B4-11.
5. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

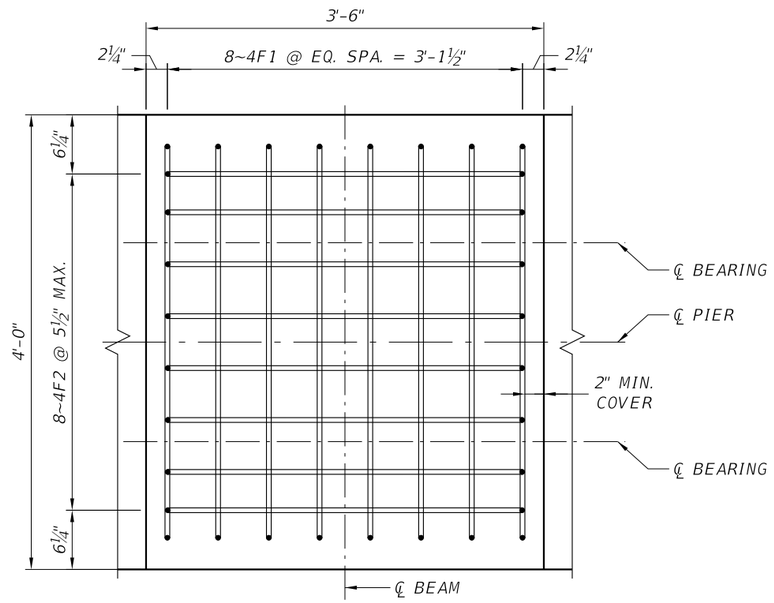
3/4" = 1'-0"

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS						ENGINEER OF RECORD:	DRAWN BY:	STATE OF FLORIDA DEPARTMENT OF TRANSPORTATION			SHEET TITLE: PIER DETAILS (1 OF 6)	REF. DWG. NO.
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION		CHECKED BY:					
							DESIGNED BY:	ROAD NO.	COUNTY	FINANCIAL PROJECT ID	PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE – B4	SHEET NO. B4-6
							CHECKED BY:					

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

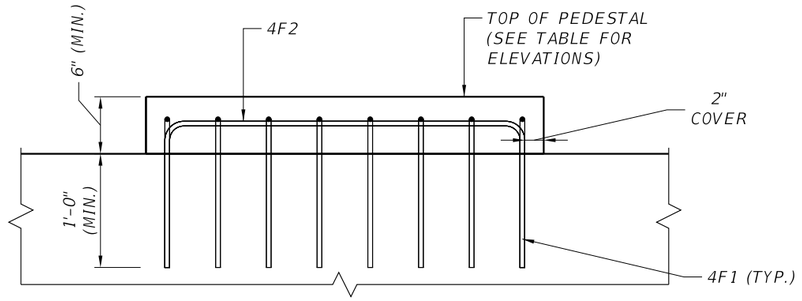
FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)



PEDESTAL PLAN

(BEAM LINE 1 SHOWN; BAR SPACING MAY BE ADJUSTED TO CLEAR THE CAP REINFORCEMENT)

DIRECTION OF STATIONING



PEDESTAL ELEVATION

TOP OF PEDESTAL ELEVATIONS	
BEAM NO.	ELEVATION (FEET)
1-1 / 2-1	44.267
1-2 / 2-2	44.267

NOTES:

1. FOR LOCATION OF THE PEDESTALS, SEE SHEET B4-5; FOR SECTIONS A-A, B-B AND C-C, SEE SHEET B4-6.
2. PEDESTALS AT THE OTHER BEAM LINES ARE SIMILAR.
3. PEDESTALS ARE CAST ON THE CAP; THE CAP TOP IS A CONSTRUCTION JOINT.
4. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

3/4" = 1'-0"

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS					
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DRAWN BY:
CHECKED BY:
DESIGNED BY:
CHECKED BY:

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

ROAD NO.	COUNTY	FINANCIAL PROJECT ID

SHEET TITLE: PIER DETAILS (2 OF 6)

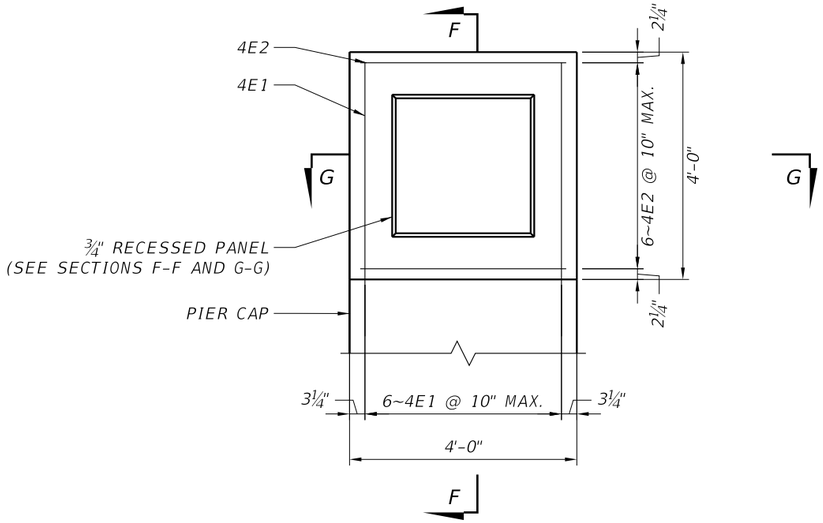
PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE - B4

REF. DWG. NO.

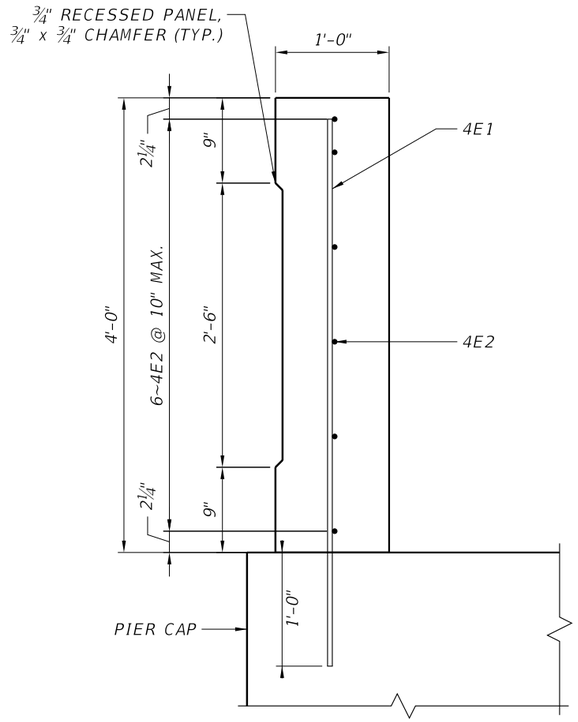
SHEET NO. B4-7

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

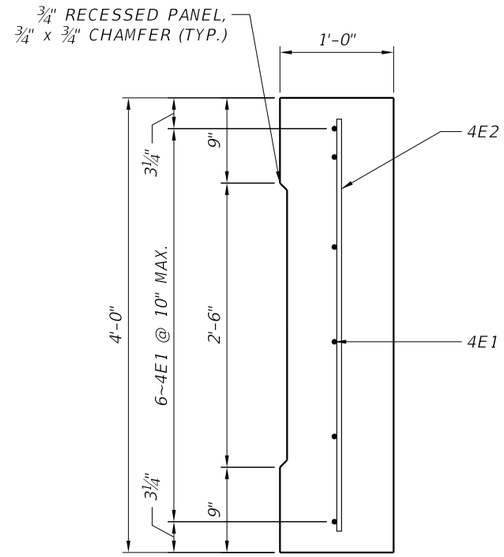
FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)



VIEW E-E
(LEFT CHEEK WALL)



SECTION F-F
(THROUGH CHEEK WALL)



SECTION G-G
(THROUGH CHEEK WALL)

NOTES:

1. FOR LOCATION OF VIEW E-E, SEE SHEET B4-5 (PLAN). SECTIONS F-F AND G-G ARE LOCATED IN VIEW E-E.
2. VIEW E-E AND SECTIONS F-F AND G-G SHOW THE LEFT CHEEK WALL; THE RIGHT CHEEK WALL IS SIMILAR (MIRRORED). THE RECESSED PANEL IS ON THE OUTSIDE FACE OF EACH CHEEK WALL; THE INSIDE FACE IS FLAT.
3. CAP REINFORCEMENT IS NOT SHOWN IN SECTION F-F FOR CLARITY.
4. CHEEK WALLS ARE CAST ON THE CAP; THE CAP TOP IS A CONSTRUCTION JOINT.
5. CHEEK WALL VERTICALS 4E1 ARE EMBEDDED 1'-0" (MIN.) INTO THE CAP, STRAIGHT; SEE SECTION F-F.
6. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

3/8" = 1'-0" 3/4" = 1'-0"

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS					
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DRAWN BY:
CHECKED BY:
DESIGNED BY:
CHECKED BY:

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

ROAD NO.	COUNTY	FINANCIAL PROJECT ID

SHEET TITLE: PIER DETAILS (3 OF 6)

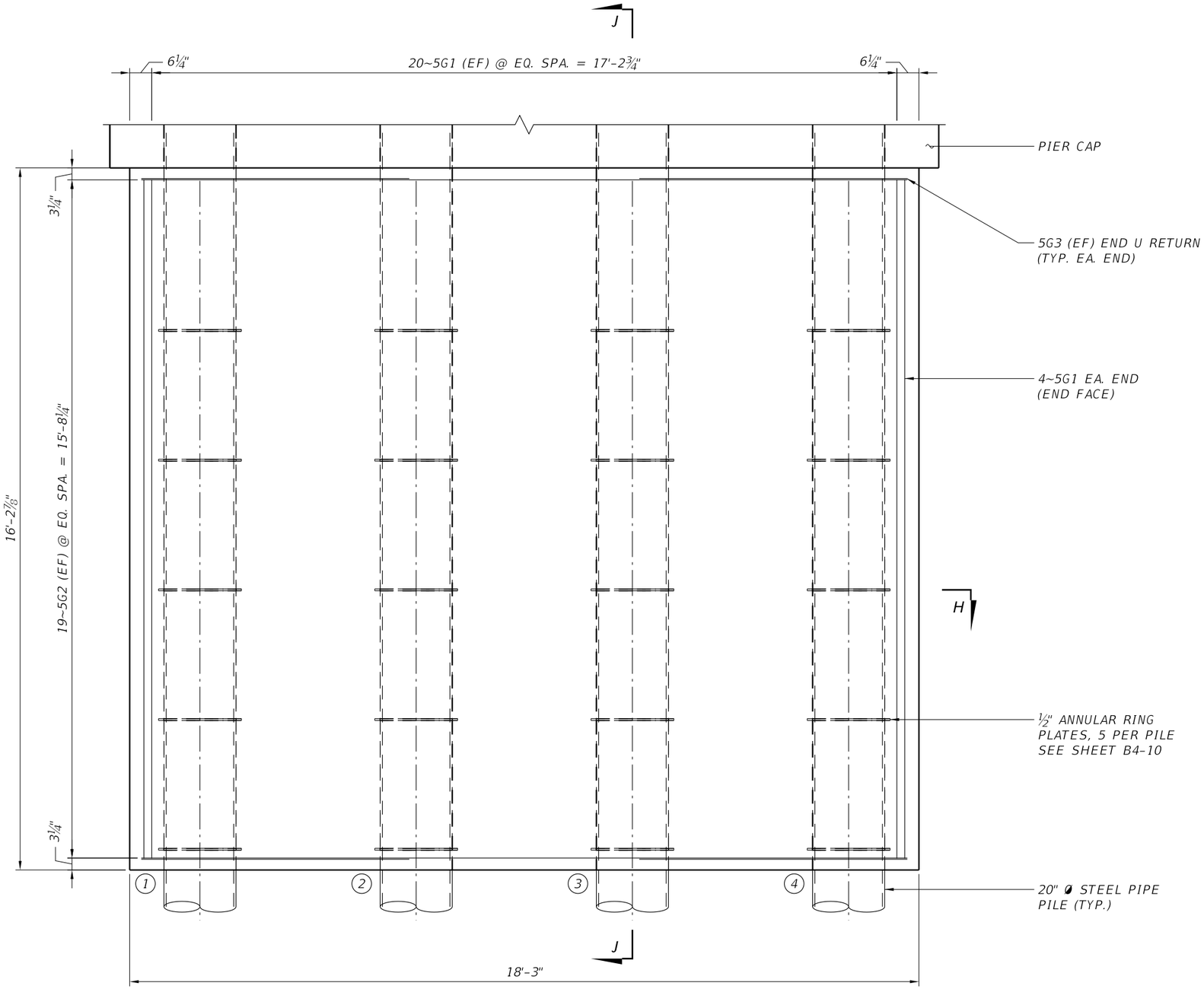
PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE - B4

REF. DWG. NO.

SHEET NO. B4-8

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)



PIER WALL ELEVATION

(LOOKING AHEAD STATION)

3/8" = 1'-0"

NOTES:

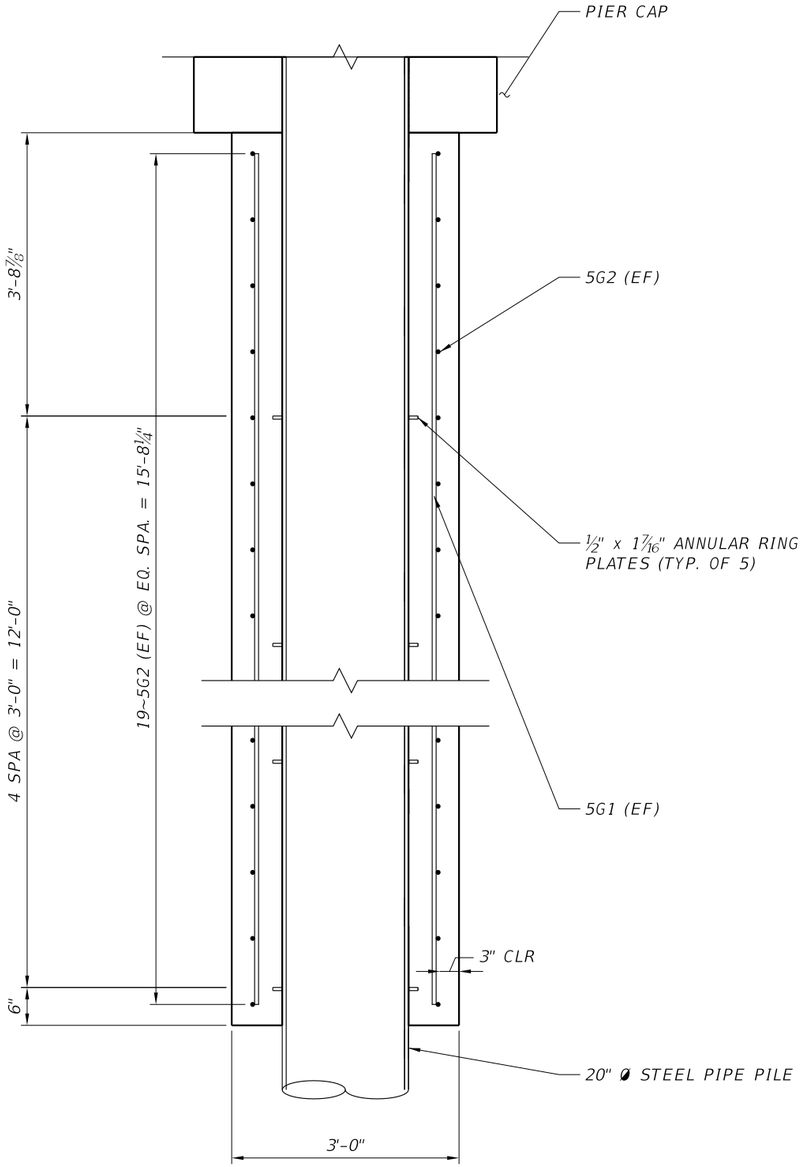
1. FOR THE LOCATION OF THE PIER WALL, SEE SHEET B4-5 (PLAN AND ELEVATION). SECTIONS H-H AND J-J ARE LOCATED IN THE PIER WALL ELEVATION; FOR SECTION J-J AND THE ANNULAR RING PLATE DETAILS, SEE SHEET B4-10; FOR SECTION H-H, SEE SHEET B4-11.
2. PIER WALL REINFORCEMENT: #5 BARS AT 11" MAXIMUM EACH WAY ON EACH FACE (5G1 VERTICALS, 5G2 HORIZONTALS), 3" CLEAR COVER; 5G3 END U RETURNS AT EACH HORIZONTAL WITH 6'-0" CONTACT LAP LEGS ALONG THE FACES; 4~5G1 VERTICALS IN EACH END FACE. THE HORIZONTALS ARE THE OUTER LAYER.
3. REINFORCEMENT IS SHOWN REPRESENTATIVELY: THE FIRST AND LAST BAR OF EACH RUN ARE DRAWN.
4. ANNULAR RING PLATES: 20 PLATES, ½" THICK, 23" O.D. x 20⅛" I.D., 5 PER PILE WITH THE PLATE TOPS 6", 3'-6", 6'-6", 9'-6", 12'-6" ABOVE THE BOTTOM OF THE WALL, CENTERED ON THE PILES AND WELDED TO THEM WITH TWO CONTINUOUS 5⁄16" FILLETS (TOP AND BOTTOM OF EACH PLATE); SEE SHEET B4-10. THE PLATES ARE SHOWN HIDDEN IN THE ELEVATION.
5. PILES ARE SHOWN AT THEIR NOMINAL LOCATIONS; THE RING PLATES ARE SIZED FOR A 3" RADIAL PILE PLACEMENT TOLERANCE.
6. THE CAP IS SHOWN BROKEN 1'-0" ABOVE ITS BOTTOM; FOR THE CAP, SEE SHEETS B4-5 AND B4-6.
7. MODEL CHECK (REMOVE BEFORE ISSUE): THE SELECTED RING CALCULATION (PIER-WALL-CASE, D-078, 22 OCTOBER 2026) REPORTS THE FAILED CHECK RING-GRID (RING CLEARANCE TO INNER FACE-GRID LAYER: 1.125 IN REQUIRED, -0.75 IN AVAILABLE); RESOLVE BEFORE ISSUE.
8. MODEL CHECK (REMOVE BEFORE ISSUE): THE PIER WALL VERTICALS STOP 3⅜" BELOW THE CAP AND 3⅜" ABOVE THE BOTTOM OF THE WALL IN THE CAD MODEL (NO ANCHORAGE INTO THE CAP); DETAIL THE WALL-TO-CAP CONNECTION AND ADD IT TO THE MODEL.
9. MODEL CHECK (REMOVE BEFORE ISSUE): CAD MODEL STATUS OF THE PIER WALL STEEL: "CAD RING PLATES 1/2 IN MATCH THE SELECTED D-078 CASE; WALL DIMENSIONS MATCH D-075 WITHIN 0.001 IN; RING-GRID TOLERANCE FAILURE REMAINS; ENGINEERING ACCEPTANCE OPEN".
10. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS						ENGINEER OF RECORD:	DRAWN BY:	STATE OF FLORIDA DEPARTMENT OF TRANSPORTATION			SHEET TITLE: PIER DETAILS (4 OF 6)	REF. DWG. NO.
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION		CHECKED BY:					
							DESIGNED BY:	ROAD NO.	COUNTY	FINANCIAL PROJECT ID	PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE – B4	SHEET NO. B4-9
							CHECKED BY:					

FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)

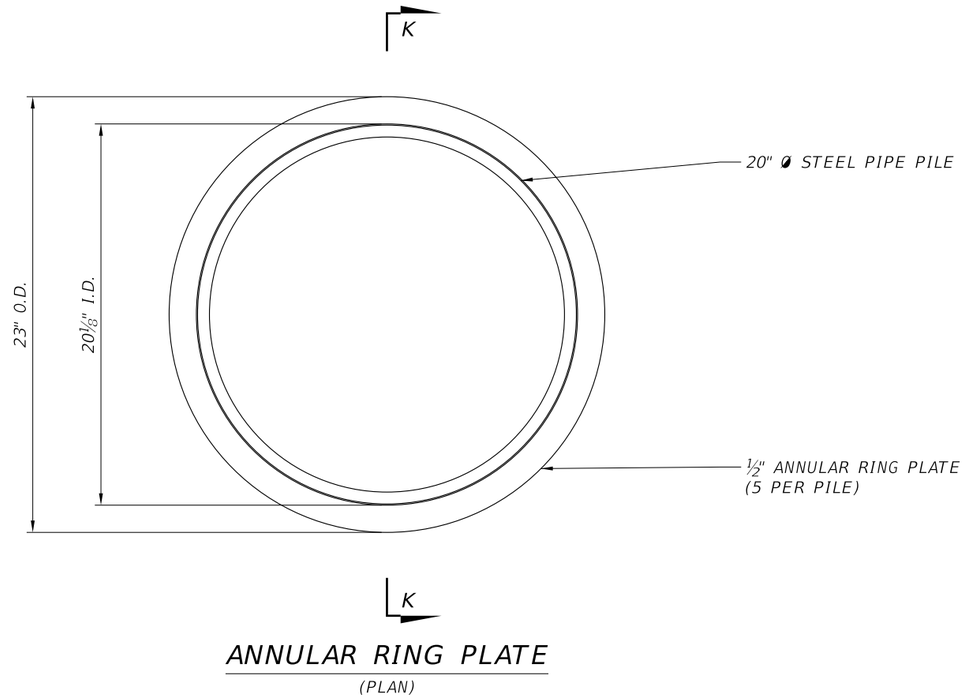
THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.



SECTION J-J

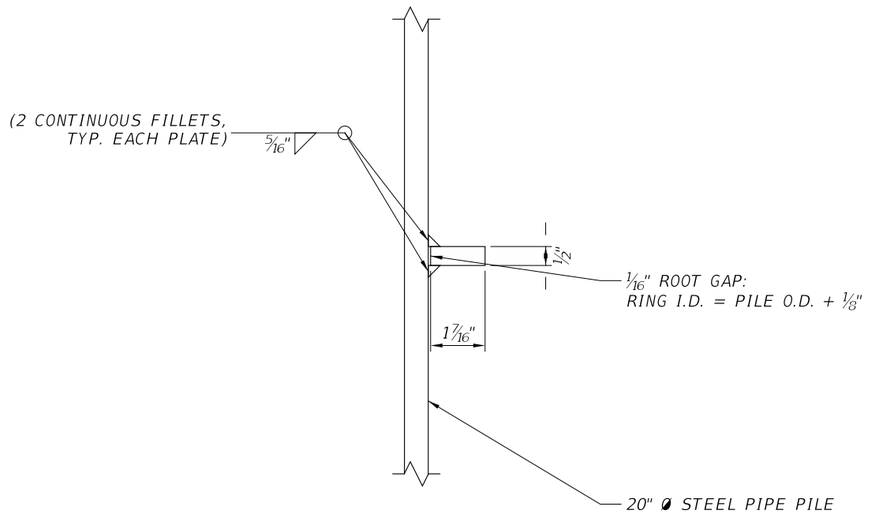
(THROUGH PIER WALL AT PILE 3)

1/2" – 1'-0"



ANNULAR RING PLATE

(PLAN)



SECTION K-K

(THROUGH ANNULAR RING PLATE)

2" – 1'-0"

NOTES:

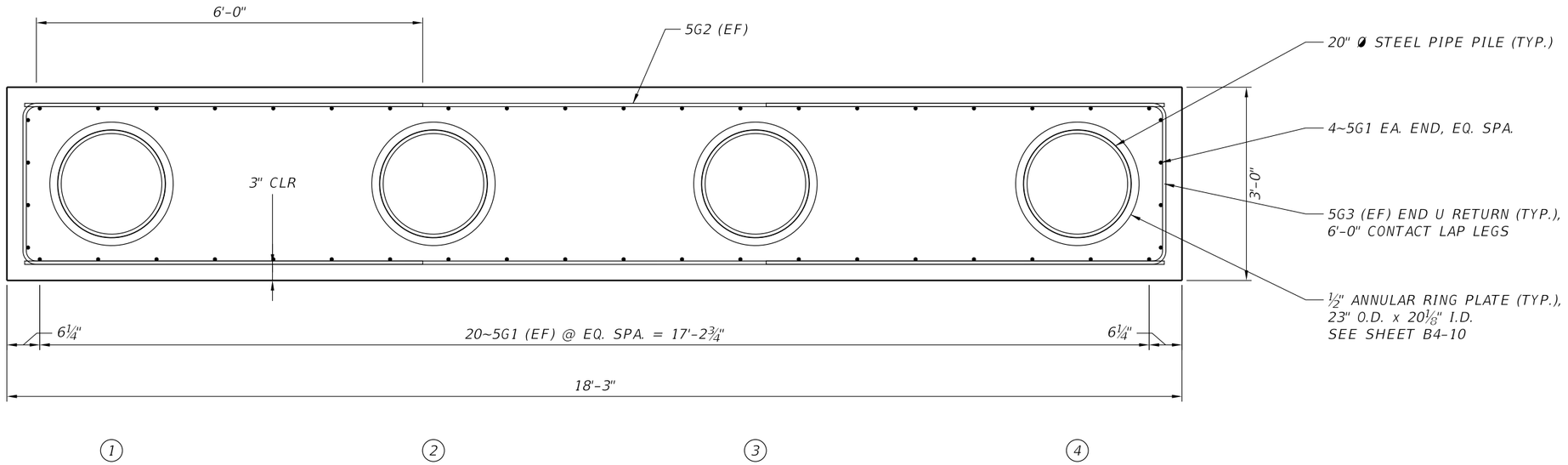
1. FOR THE LOCATION OF SECTION J-J, SEE THE PIER WALL ELEVATION, SHEET B4-9. SECTION K-K IS LOCATED IN THE ANNULAR RING PLATE PLAN.
2. SECTION J-J IS BROKEN BETWEEN RING ROWS 2 AND 4; THE RING PLATE TOPS ARE DIMENSIONED FROM THE BOTTOM OF THE WALL AND THE HORIZONTAL BAR SERIES IS GIVEN IN FULL.
3. ALL BARS ARE SHOWN IN SECTION J-J; BARS IN THE SECTION PLANE ARE THE NEAREST BAR OF THEIR SERIES WITHIN 1'-3" BEYOND THE CUT. THE END U RETURNS AND THE END-FACE VERTICALS LIE BEYOND THE CUT AND ARE NOT SHOWN; SEE SECTION H-H, SHEET B4-11.
4. ANNULAR RING PLATES: ½" THICK, 23" O.D. x 20⅛" I.D. (1 7/16" RADIAL WIDTH), Fy = 50 KSI, Fu = 65 KSI; WELDED TO THE PIPE PILE WITH TWO CONTINUOUS 5/16" FILLET WELDS, E70XX ELECTRODES, ONE AT THE TOP AND ONE AT THE BOTTOM OF EACH PLATE. THE 1/16" RADIAL ROOT GAP IS AN ANALYSIS ALLOWANCE; THE WPS SHALL ACCEPT IT.
5. PIPE PILES ARE SHOWN WITH THEIR NOMINAL 20" OUTSIDE DIAMETER; THE WELD SOLIDS ARE NOT IN THE CAD MODEL.
6. THE PLATE AND WELD SIZES ARE THOSE OF THE SELECTED PIER WALL RING CALCULATION (PIER-WALL-CASE, D-078, 22 OCTOBER 2026).
7. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS						ENGINEER OF RECORD:	DRAWN BY:	STATE OF FLORIDA DEPARTMENT OF TRANSPORTATION				SHEET TITLE: PIER DETAILS (5 OF 6)	REF. DWG. NO.
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION		CHECKED BY:						
							DESIGNED BY:	ROAD NO.	COUNTY	FINANCIAL PROJECT ID		PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE – B4	SHEET NO. B4-10
							CHECKED BY:						

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)



SECTION H-H

(HORIZONTAL SECTION THROUGH PIER WALL AT RING ROW 3)

NOTES:

1. FOR THE LOCATION OF SECTION H-H, SEE THE PIER WALL ELEVATION, SHEET B4-9.
2. ALL BARS ARE SHOWN IN SECTION H-H; BARS IN THE SECTION PLANE ARE THE NEAREST BAR OF THEIR SERIES WITHIN 1'-3" BELOW THE CUT. THE CUT IS THROUGH RING ROW 3; THE RING PLATES AND THE PIPE PILES ARE SHOWN CUT.
3. THE END U RETURNS LAP THE HORIZONTALS ALONG BOTH FACES; THE U LEG LENGTH IS DIMENSIONED OVER THE BACK FACE AT THE LEFT END AND IS THE SAME AT EVERY END U RETURN.
4. COVER IS DIMENSIONED TO THE OUTERMOST BAR (THE HORIZONTALS) AT THE FRONT FACE; IT IS 3" ON EVERY FACE.
5. PILE NUMBERS ARE THOSE OF SHEET B4-5.
6. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

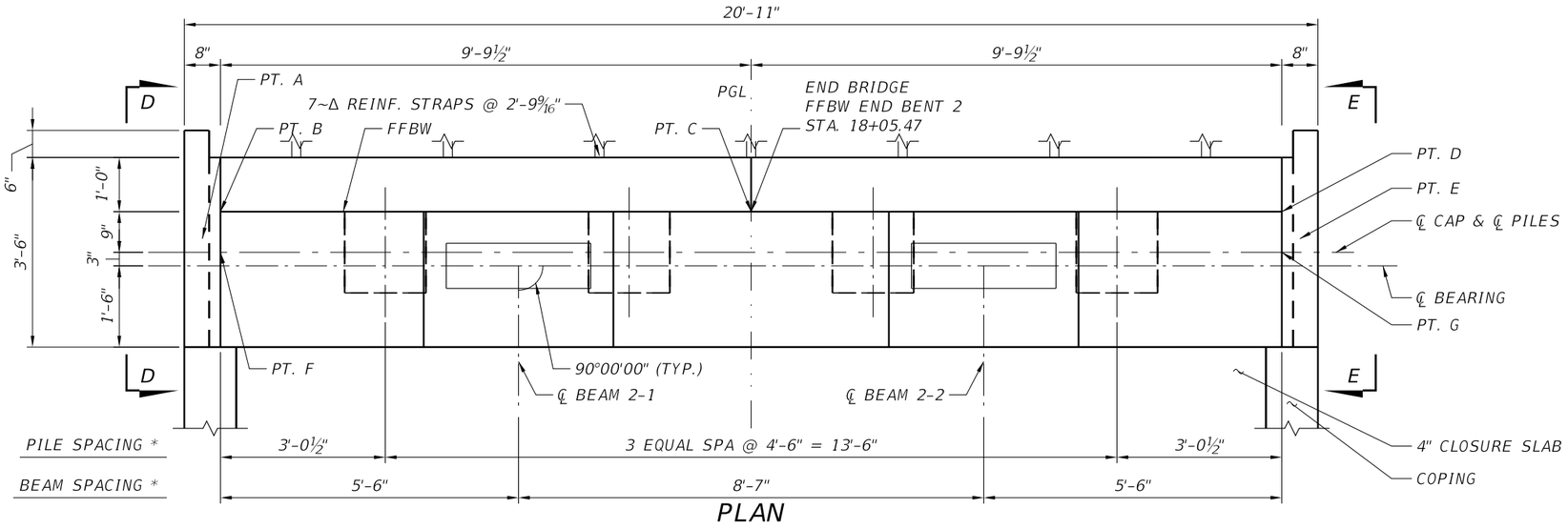
1/2" = 1'-0"

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

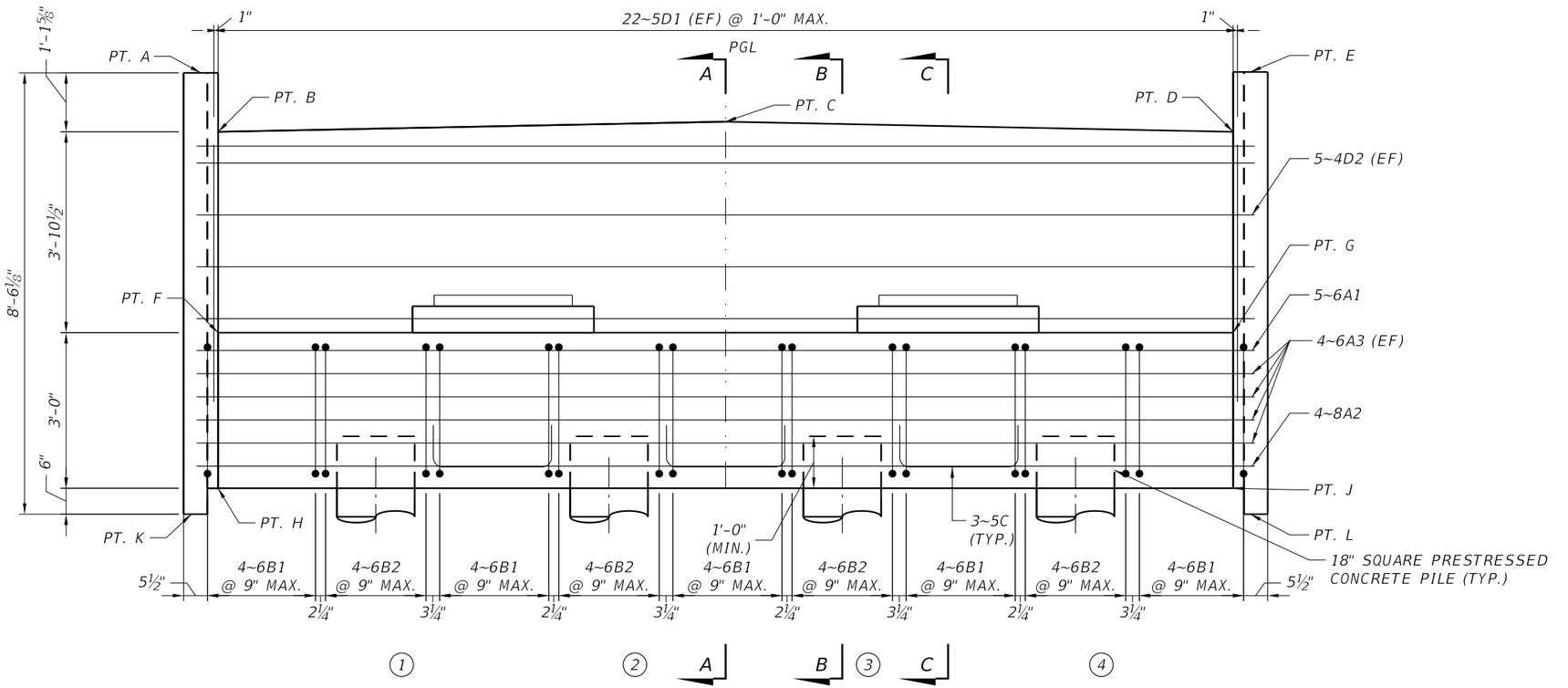
REVISIONS						ENGINEER OF RECORD:	DRAWN BY:	STATE OF FLORIDA DEPARTMENT OF TRANSPORTATION			SHEET TITLE: PIER DETAILS (6 OF 6)	REF. DWG. NO.
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION		CHECKED BY:					
							DESIGNED BY:	ROAD NO.	COUNTY	FINANCIAL PROJECT ID	PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE - B4	SHEET NO. B4-11
							CHECKED BY:					

FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.



PLAN



* DIMENSIONS MEASURED ALONG FFBW

ELEVATION

(LOOKING AHEAD STATION)

3/8" = 1'-0"

DIRECTION OF STATIONING

TABLE OF ELEVATIONS	
POINT	ELEVATION
A	48.260
B	47.123
C	47.319
D	47.123
E	48.277
F	43.250
G	43.250
H	40.250
J	40.250
K	39.750
L	39.750

PILE CUTOFF ELEVATIONS		
PILE NO.	CUTOFF EL.	PGL OFFSET
1	41.2	-6'-9"
2	41.2	-2'-3"
3	41.2	2'-3"
4	41.2	6'-9"

NOTES:

1. STIRRUP SPACING IS DIMENSIONED TO BAR CENTERLINES.
2. REINFORCEMENT IS SHOWN REPRESENTATIVELY: THE FIRST AND LAST BAR OF EACH RUN ARE DRAWN.
3. FOR SECTIONS A-A AND B-B AND DETAILS "A" AND "B", SEE SHEET B4-13. FOR SECTION C-C AND VIEWS D-D AND E-E, SEE SHEET B4-14. FOR SECTIONS F-F AND G-G AND PEDESTAL DETAILS, SEE SHEET B4-15.
4. FOR THE SOIL REINFORCEMENT (Δ) LEGEND, SEE SHEET B4-13.
5. FOR REINFORCING BAR LIST, SEE SHEET B4-16.
6. PILE OFFSETS ARE FROM THE PGL, + TO THE RIGHT LOOKING UP-STATION. CUTOFF ELEVATIONS ARE TO THE NEAREST 0.1 FT.

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS					
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DRAWN BY:	
CHECKED BY:	
DESIGNED BY:	
CHECKED BY:	

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

ROAD NO.	COUNTY	FINANCIAL PROJECT ID

SHEET TITLE: END BENT 2

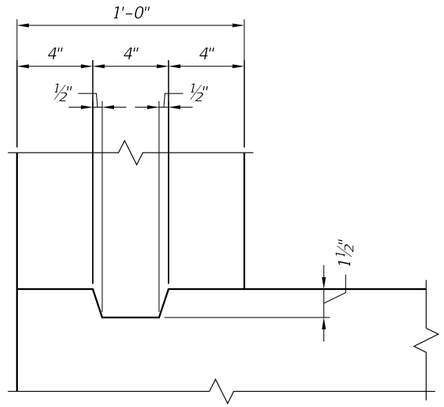
PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE - B4

REF. DWG. NO.

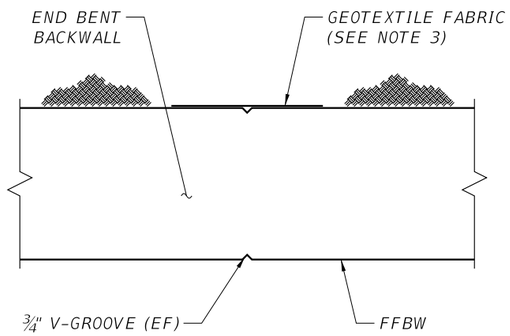
SHEET NO. B4-12

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)



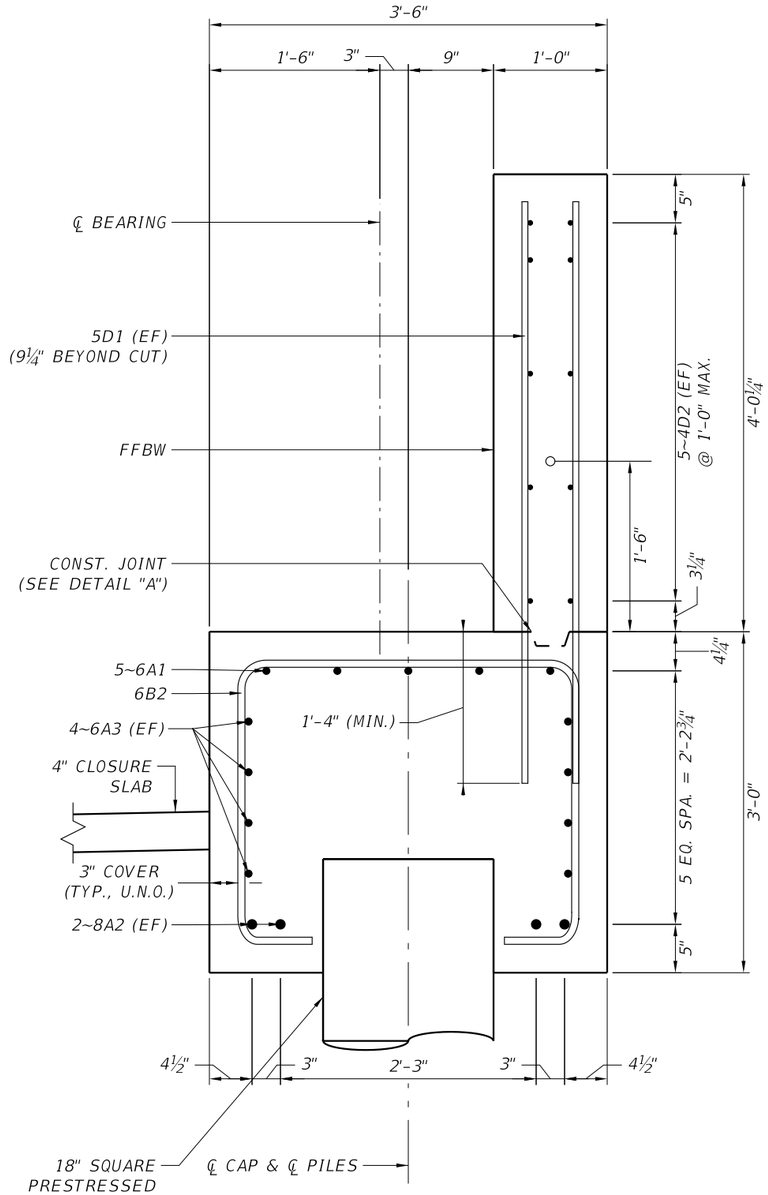
DETAIL "A"
(CONSTRUCTION JOINT)



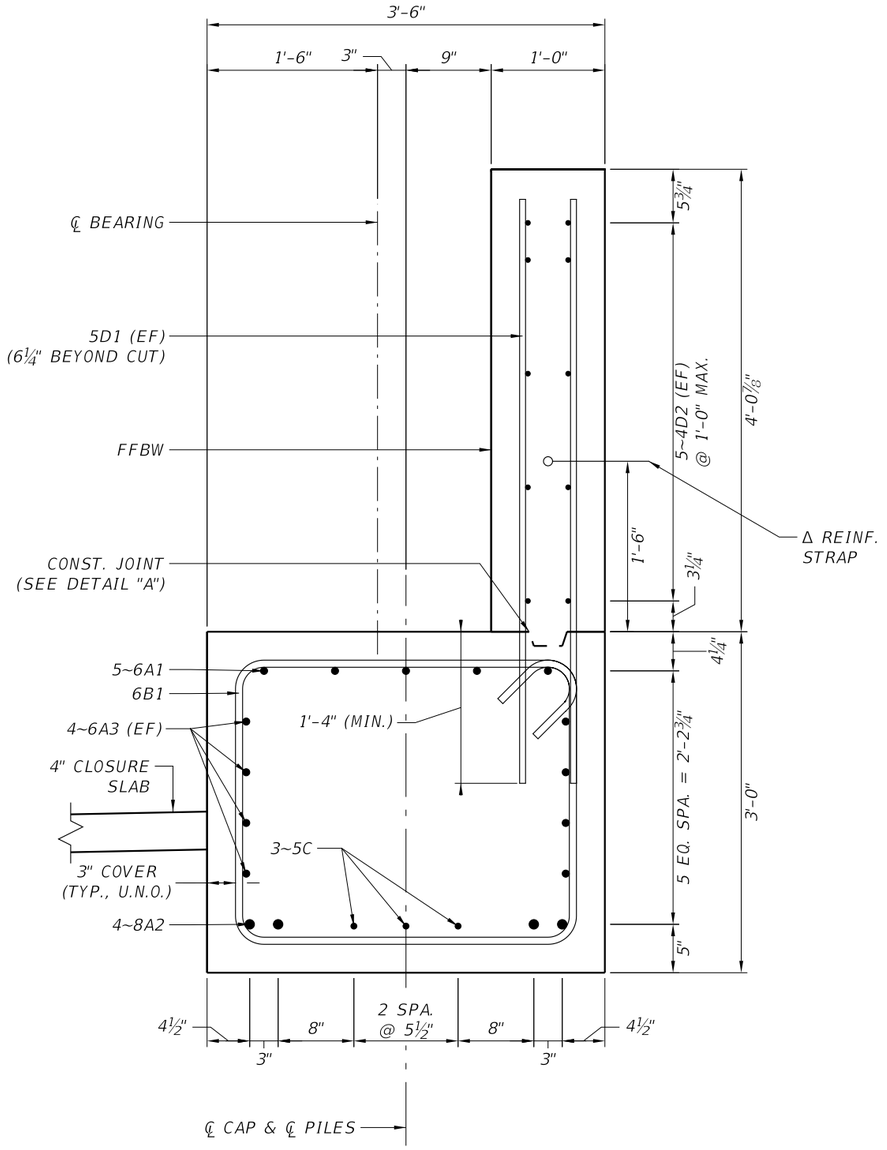
DETAIL "B"
(END BENT BACKWALL V-GROOVE)

LEGEND:

Δ REINFORCING STRAPS AND ATTACHMENTS FOR THE END BENT SHALL BE DESIGNED AND FURNISHED BY THE PROPRIETARY WALL PROVIDER. THE STRAPS SHALL BE DESIGNED TO RESIST A SERVICE LOAD OF 3.09 K/FT AND A FACTORED LOAD OF 3.74 K/FT (STRENGTH I) PERPENDICULAR TO THE END BENT BACKWALL.



SECTION B-B
(SECTION AT PILES)



SECTION A-A
(SECTION BETWEEN PILES)

NOTES:

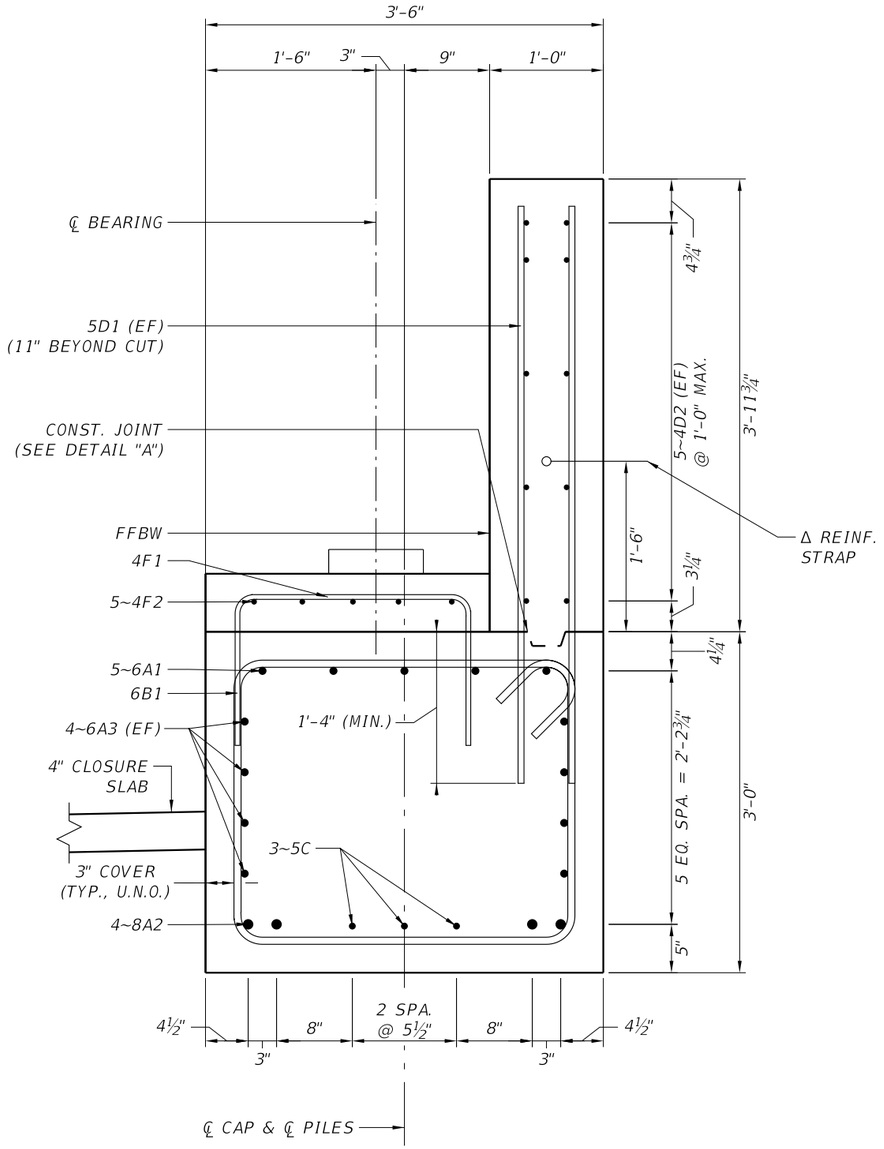
1. FOR LOCATION OF SECTIONS, SEE SHEET B4-12.
2. ALL BARS ARE SHOWN IN SECTIONS; BARS IN THE SECTION PLANE ARE THE NEAREST BAR OF THEIR SERIES WITHIN 1'-3" BEYOND THE CUT.
3. ATTACH TYPE D-5 (SPECIFICATIONS SECTION 985) GEOTEXTILE FABRIC, 1'-0" WIDE AND FULL HEIGHT OF FILL, TO THE BACK OF BACKWALL AT V-GROOVE LOCATIONS WITH AN ADHESIVE APPROVED BY THE ENGINEER.
4. MODEL CHECK (REMOVE BEFORE ISSUE): DETAILS "A" AND "B" ARE SCHEMATIC; THE CONSTRUCTION JOINT KEY AND THE V-GROOVE LOCATIONS ARE NOT IN THE CAD MODEL.
5. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

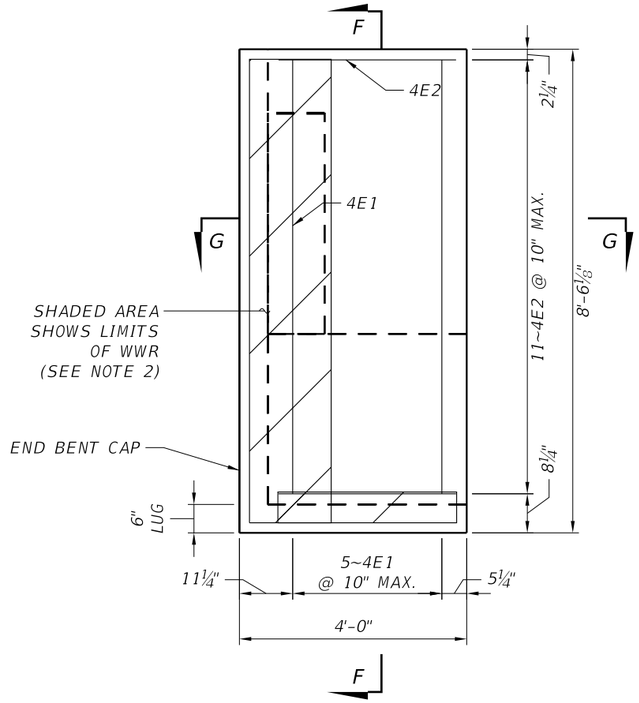
REVISIONS						ENGINEER OF RECORD:	DRAWN BY:	STATE OF FLORIDA DEPARTMENT OF TRANSPORTATION			SHEET TITLE: END BENT 2 DETAILS (1 OF 3)	REF. DWG. NO.
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION		CHECKED BY:					
							DESIGNED BY:	ROAD NO.	COUNTY	FINANCIAL PROJECT ID	PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE – B4	SHEET NO. B4-13
							CHECKED BY:					

FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)

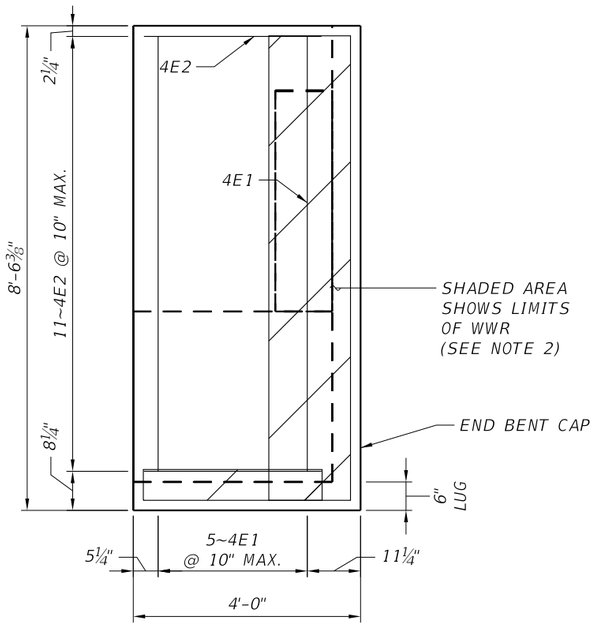
THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.



SECTION C-C
(SECTION AT CENTERLINE OF BEAM)



VIEW D-D
(LEFT CHEEK WALL)



VIEW E-E
(RIGHT CHEEK WALL)

NOTES:

1. FOR LOCATION OF SECTION C-C AND VIEWS D-D AND E-E, SEE SHEET B4-12.
2. WELDED WIRE REINFORCEMENT (WWR) 4x4-W1.4xW1.4 IN THE CHEEK LUGS SHALL BE TIED TO THE END BENT REINFORCING STEEL.
3. FOR CHEEK TOP ELEVATIONS, SEE THE TABLE OF ELEVATIONS, SHEET B4-12.
4. SECTIONS F-F AND G-G ARE LOCATED IN VIEW D-D; FOR SECTIONS F-F AND G-G, SEE SHEET B4-15.
5. FOR THE Δ REINFORCING STRAP LEGEND, SEE SHEET B4-13.
6. MODEL CHECK (REMOVE BEFORE ISSUE): THE BACKWALL TOP IS LEVEL ACROSS THE WALL THICKNESS IN THE CAD MODEL (ITS SLOPE RUNS ALONG THE BENT TO THE CROWN ONLY); THE SECTIONS SHOW THE MODEL. ADD THE CROSS SLOPE TO THE MODEL.
7. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

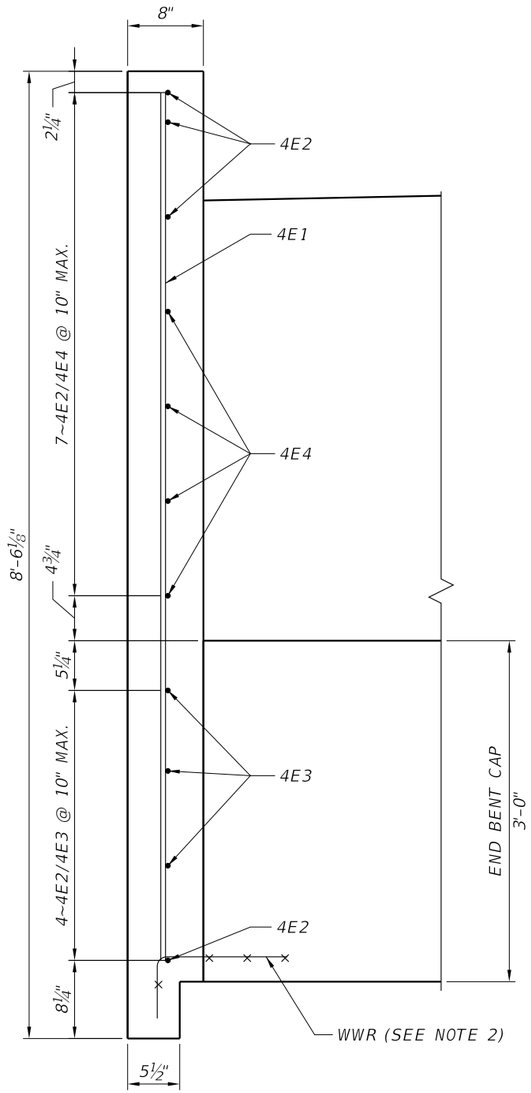
3/4" = 1'-0" 3/8" = 1'-0"

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS						ENGINEER OF RECORD:	DRAWN BY:	STATE OF FLORIDA DEPARTMENT OF TRANSPORTATION			SHEET TITLE: END BENT 2 DETAILS (2 OF 3)	REF. DWG. NO.
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION		CHECKED BY:					
							DESIGNED BY:	ROAD NO.	COUNTY	FINANCIAL PROJECT ID	PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE – B4	SHEET NO. B4-14
							CHECKED BY:					

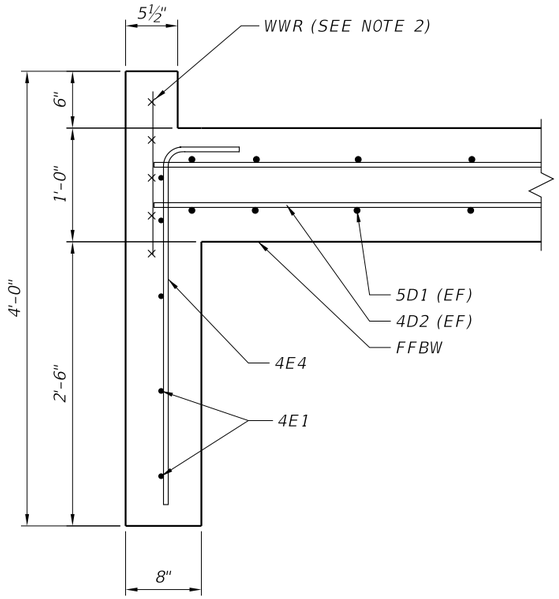
FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.



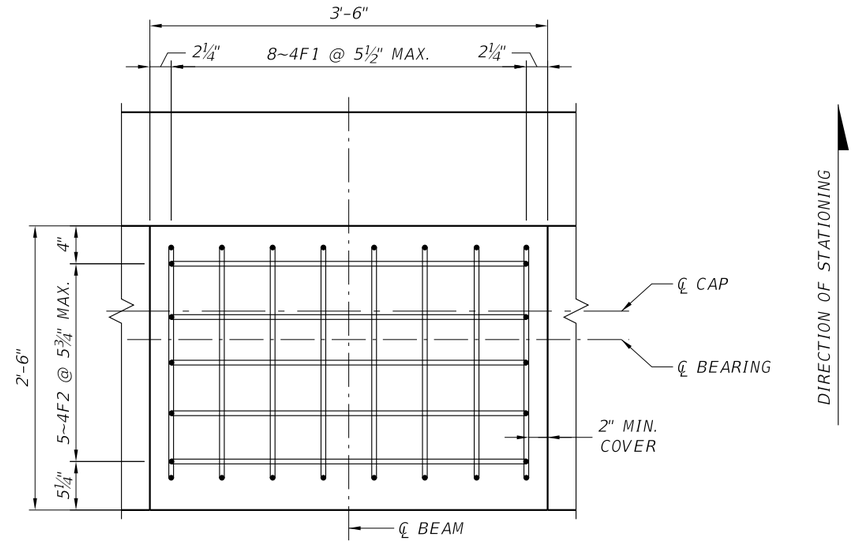
SECTION F-F

(THROUGH CHEEK WALL)



SECTION G-G

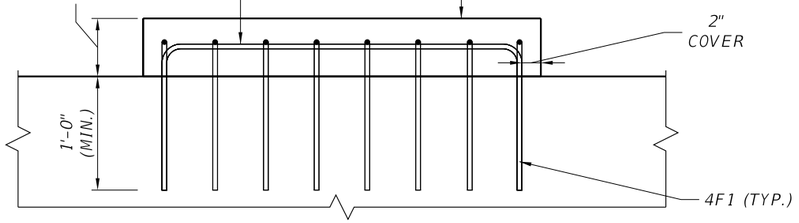
(THROUGH CHEEK WALL AND BACKWALL)



PEDESTAL PLAN

(BEAM LINE 1 SHOWN; BAR SPACING MAY BE ADJUSTED TO CLEAR THE CAP REINFORCEMENT)

TOP OF PEDESTAL ELEVATIONS	
BEAM NO.	ELEVATION (FEET)
2-1	43.760
2-2	43.760



PEDESTAL ELEVATION

NOTES:

1. FOR LOCATION OF SECTIONS F-F AND G-G, SEE VIEW D-D, SHEET B4-14; SECTIONS F-F AND G-G SHOW THE LEFT CHEEK WALL, THE RIGHT CHEEK WALL IS SIMILAR (MIRRORED).
2. WELDED WIRE REINFORCEMENT (WWR) 4x4-W1.4xW1.4 IN THE CHEEK LUGS SHALL BE TIED TO THE END BENT REINFORCING STEEL.
3. CAP AND BACKWALL REINFORCEMENT IS NOT SHOWN IN SECTION F-F FOR CLARITY.
4. PEDESTALS AT THE OTHER BEAM LINES ARE SIMILAR.
5. PEDESTALS ARE CAST ON THE CAP; THE CAP TOP IS A CONSTRUCTION JOINT.
6. FOR REINFORCING BAR LIST, SEE SHEET B4-16.

3/4" = 1'-0"

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS					
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DRAWN BY:
CHECKED BY:
DESIGNED BY:
CHECKED BY:

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

ROAD NO.	COUNTY	FINANCIAL PROJECT ID

SHEET TITLE: END BENT 2 DETAILS (3 OF 3)

PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE - B4

REF. DWG. NO.

SHEET NO. B4-15

FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

MARK		LENGTH	NO	TYP	STY		B	C	D	E	F	H	J	K	N	⌀
SIZE	DES	FT IN	BARS	BAR	A	G	FT IN	FT IN	FT IN	FT IN	FT IN	FT IN	FT IN	FT IN	NO	ANG
END BENT 1 (1 REQUIRED)																
6	A1	20-5	5	1			20-5									
8	A2	20-5	4	1			20-5									
6	A3	20-5	8	1			20-5									
6	B1	11-10	20	4	5	5	2-6	3-0								
6	B2	9-4	16	5			2-6	3-0	1-0	1-0						
5	C	3-9	9	11			2-4	0-10	0-10							
5	D1	4-11 TO 5-2	44	1			4-11 TO 5-2									
4	D2	20-5	10	1			20-5									
4	E1	7-8	10	1			7-8									
4	E2	3-2	6	1			3-2									
4	E3	3-11	6	11			2-10	0-8	0-8							
4	E4	3-8	10	10			3-1¾	0-8								
4	F1	4-6	16	11			2-0¾	1-4	1-4							
4	F2	5-6	10	11			3-2	1-3½	1-3½							
END BENT 2 (1 REQUIRED)																
6	A1	20-5	5	1			20-5									
8	A2	20-5	4	1			20-5									
6	A3	20-5	8	1			20-5									
6	B1	11-10	20	4	5	5	2-6	3-0								
6	B2	9-4	16	5			2-6	3-0	1-0	1-0						
5	C	3-9	9	11			2-4	0-10	0-10							
5	D1	4-11 TO 5-2	44	1			4-11 TO 5-2									
4	D2	20-5	10	1			20-5									
4	E1	7-8	10	1			7-8									
4	E2	3-2	8	1			3-2									
4	E3	3-11	6	11			2-10	0-8	0-8							
4	E4	3-8	8	10			3-1¾	0-8								
4	F1	4-6	16	11			2-0¾	1-4	1-4							
4	F2	5-6	10	11			3-2	1-3½	1-3½							

NOTE:
FOR BAR BENDING DETAILS, REFER TO FDOT STANDARD PLANS INDEX 415-001.

MARK		LENGTH	NO	TYP	STY		B	C	D	E	F	H	J	K	N	⌀
SIZE	DES	FT IN	BARS	BAR	A	G	FT IN	FT IN	FT IN	FT IN	FT IN	FT IN	FT IN	FT IN	NO	ANG
PIER (1 REQUIRED)																
6	A1	18-8	5	1			18-8									
9	A2	18-8	4	1			18-8									
6	A3	18-8	8	1			18-8									
5	B1	12-8	24	4	5	5	2-6	3-6								
5	B2	9-8	24	5			2-6	3-6	0-10	0-10						
5	C	4-1	15	11			2-8	0-10	0-10							
5	D1	4-10	4	11			3-4¾	0-10	0-10							
5	D2	3-10	6	11			2-4¾	0-10	0-10							
4	E1	4-10	12	1			4-10									
4	E2	3-8	12	1			3-8									
4	F1	5-11	16	11			3-5¾	1-4	1-4							
4	F2	5-6	16	11			3-2	1-3½	1-3½							
5	G1	15-8	48	1			15-8									
5	G2	17-8	38	1			17-8									
5	G3	14-8	38	11			2-6	6-2½	6-2½							

NOTES:

1. TYP BAR AND STY ARE THE BAR TYPE AND HOOK STYLE OF STANDARD PLANS INDEX 415-001 (FY 2026-27): TYPE 1 STRAIGHT (B = LENGTH); TYPE 10 L-BAR (B = LONG LEG, C = SHORT LEG); TYPE 11 U-BAR (B = BASE, C AND D = LEGS); TYPE 5 OPEN STIRRUP (B = HEIGHT, C = WIDTH, D AND E = FEET); TYPE 4 CLOSED STIRRUP (B = HEIGHT, C = WIDTH); STYLE 5 = 135° STIRRUP HOOKS.
2. FIELD BEND OPEN STIRRUP HOOKS AS REQUIRED TO CLEAR PILE HEAD LOCATION.
3. LENGTHS ARE THE MODEL CENTERLINE LENGTHS ROUNDED TO THE NEAREST INCH; BEND DIMENSIONS ARE OUT-TO-OUT (SDM 4.3.10.C).
4. MODEL CHECK (REMOVE BEFORE ISSUE): END BENT 1 4E1, END BENT 2 4E1 ARE STRAIGHT IN THE CAD MODEL AND ARE LISTED AS TYPE 1; THEY ARE TO BE TYPE 10 L-BARS. ADD THE LEGS TO THE MODEL.
5. VARYING LENGTHS: END BENT 1 5D1: 8 @ 4'-11", 16 @ 5'-0", 16 @ 5'-1", 4 @ 5'-2". END BENT 2 5D1: 8 @ 4'-11", 16 @ 5'-0", 16 @ 5'-1", 4 @ 5'-2".
6. PILES AND THE CHEEK LUG WELDED WIRE REINFORCEMENT ARE NOT INCLUDED IN THIS LIST.
7. MODEL CHECK (REMOVE BEFORE ISSUE): THE PIER WALL BARS 5G1, 5G2, 5G3 CARRY THE CAD MODEL'S PW-IDENTIFIERS ("PW IDENTIFIERS ARE CAD INSTANCE IDENTIFIERS; DRAWING SDM MARKS PENDING"); WRITE THESE MARKS INTO THE MODEL (substructure_drawings/freecad_marks.py) BEFORE ISSUE.

BRIDGE NO. _____
B4 PEDESTRIAN BRIDGE

REVISIONS					
DATE	BY	DESCRIPTION	DATE	BY	DESCRIPTION

ENGINEER OF RECORD:

DRAWN BY:
CHECKED BY:
DESIGNED BY:
CHECKED BY:

STATE OF FLORIDA
DEPARTMENT OF TRANSPORTATION

ROAD NO.	COUNTY	FINANCIAL PROJECT ID.

SHEET TITLE: REINFORCING BAR LIST
REF. DWG. NO.
PROJECT NAME: D2 I-95 PEDESTRIAN BRIDGE – B4
SHEET NO. B4-16

THE OFFICIAL RECORD OF THIS SHEET IS THE ELECTRONIC FILE DIGITALLY SIGNED AND SEALED UNDER RULE 61G15-23.004, F.A.C.

FREECAD MODEL SHA-256 224DE2676A92A266
B4_Substructure_EndBents_Pier.dxf (substructure_drawings 2.8.0)